CE80362 : Circuit Analysis

Lecture 3:
Network Theorems.

Superposition Theorem

- **The superposition theorem** is a method that allows us to **determine the current through or the voltage across any resistor or branch** in a network.
- The **advantage of** using this approach **instead of mesh analysis or nodal analysis** is that **it is not necessary to use determinants or matrix algebra** to analyze a given circuit.
- The theorem states the following:
  - *"The total current through or voltage across a resistor or branch may be determined by summing the effects due to each independent source".*

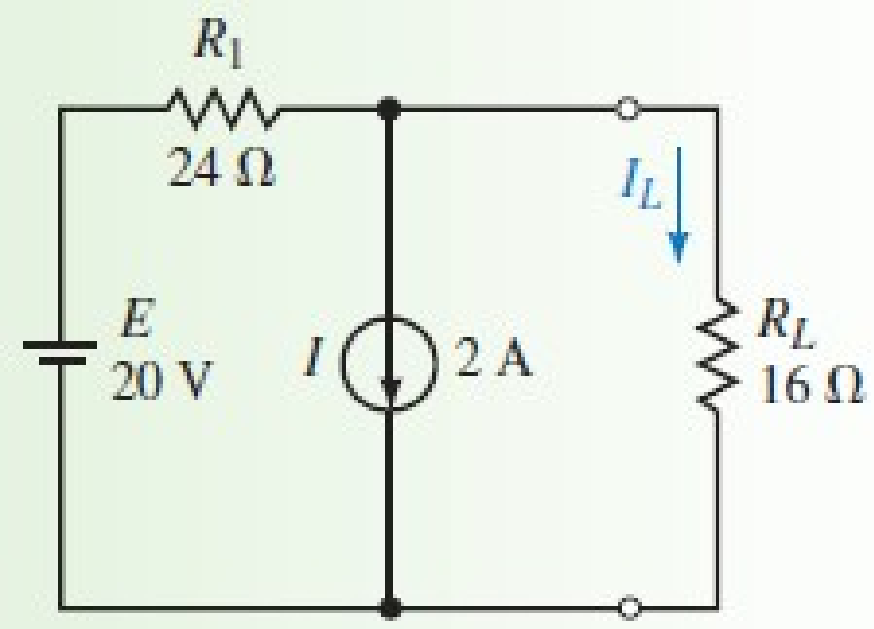
- In order to apply the superposition theorem it is necessary to:
  - **remove all sources other than the one being examined.**
- This is done by **zeroing** sources in the circuit (**only sources which are not considered**)
  - Replace a Voltage source with a short circuit, since the voltage across a short circuit ***is zero*** volts.
  - Replace a current source with an open circuit, since the current through an open circuit is zero amps.
- The **power dissipated** by any resistor can be found by:

$$P = I^2R = \frac{V^2}{R}$$

Determine the Current through any given elements.

- **Example (1):**

Determine the current in the load resistor, R_L of the circuit below. Verify that the **superposition theorem** does not apply to power.



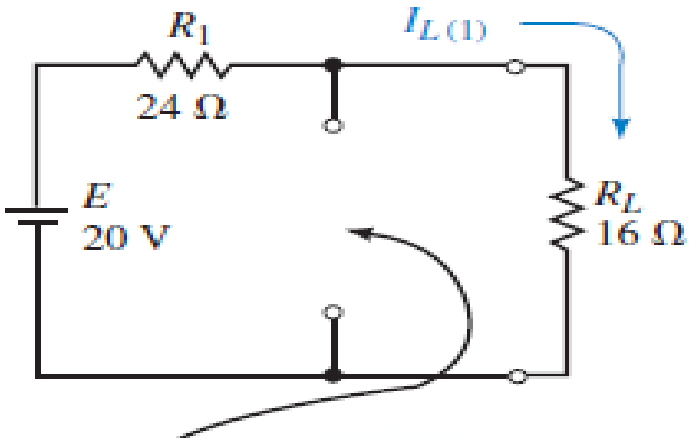
Solution:

(a) Consider Only the voltage Source:

We first determine the current through **R_L due to the voltage source** by **removing the current source** and **replacing in with an open circuit** (zero amps)

- The resulting current through $\mathbf{R_L}$ is determined from Ohm's law as:

$$I_{L(1)} = \frac{20\ \text{V}}{16\ \Omega + 24\ \Omega} = 0.500\ \text{A}$$



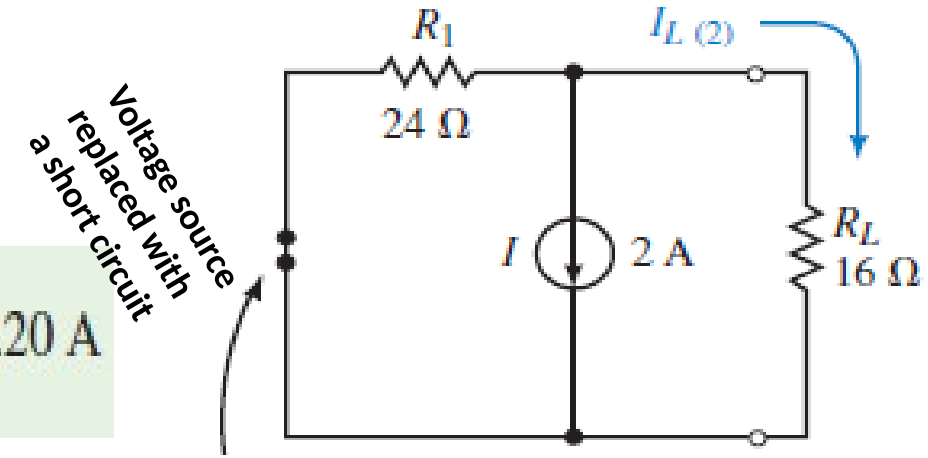
Current source replaced with an open circuit

(b) Consider the Current Source

We determine the current through ***RL*** due to the current source by **removing the voltage source** and **replacing it with a short circuit** (zero volts)

- By CDR, the resulting current through *RL* is:

$$I_{L(2)} = -\left(\frac{24\ \Omega}{24\ \Omega + 16\ \Omega}\right)(2\ \text{A}) = -1.20\ \text{A}$$



- The resultant current through R_L is found by **applying the superposition theorem**:

$$I_L = 0.5\ A + (-1.2\ A) = -0.700\ A$$

- The negative sign indicates that the current through R_L is opposite to the assumed reference direction.
- Consequently, the current through R_L will be **upward with a magnitude of 0.7 A**.

If we assume (incorrectly) that the superposition theorem applies for power, we will have the power due the first source given as

$$P_1 = I_{L(1)}^2 R_L = (0.5\ \text{A})^2(16\ \Omega) = 4.0\ \text{W}$$

and the power due the second source as

$$P_2 = I_{L(2)}^2 R_L = (1.2\ \text{A})^2(16\ \Omega) = 23.04\ \text{W}$$

The total power, if superposition applies, will be

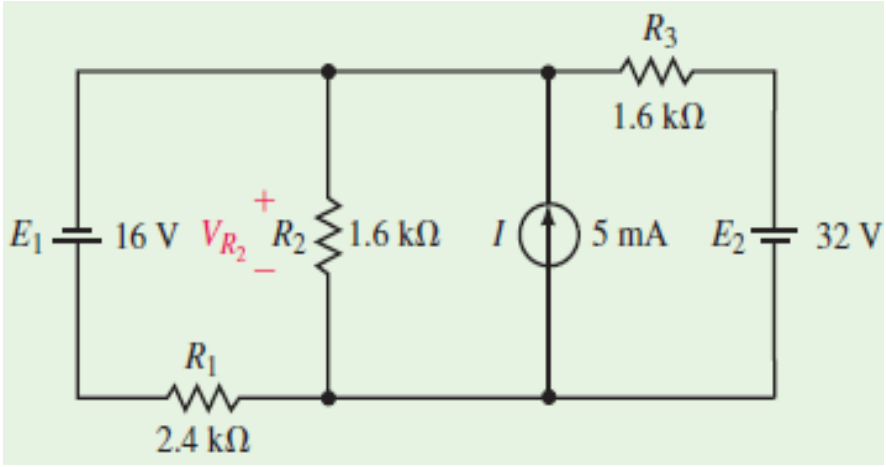
$$P_\text{T} = P_1 + P_2 = 4.0\ \text{W} + 23.04\ \text{W} = 27.04\ \text{W}$$

Clearly, this result is wrong, since the actual power dissipated by the load resistor is correctly given as

$$P_L = I_L^2 R_L = (0.7\ \text{A})^2(16\ \Omega) = 7.84\ \text{W}$$

Determine the voltage across any component.

- **Example (2):** Apply superposition theorem to determine the voltage drop across the resistor R_2 of the circuit.

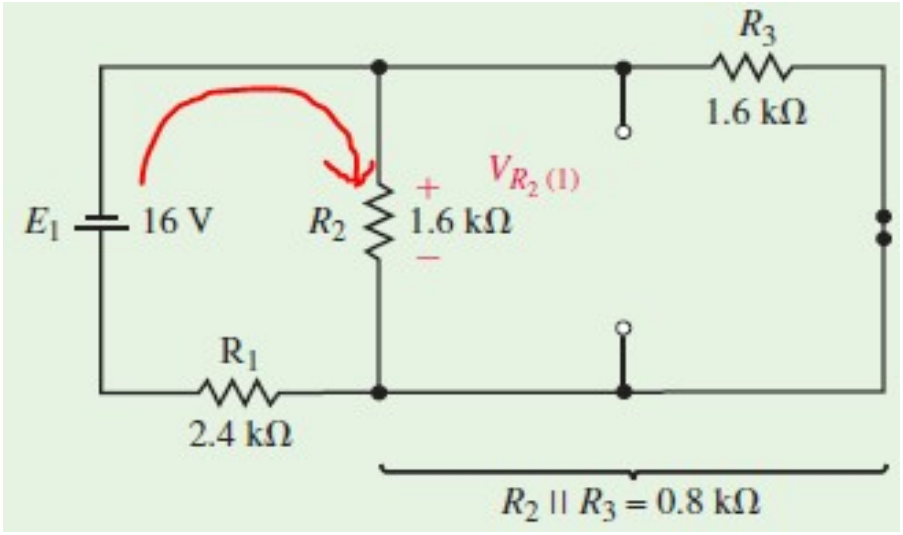


Solution:

We consider every Source individually, while others are zeroed.

(a) If ONLY *E1* is Considered:

- Since $R_2 || R_3 = 0.8k$



- By VDR, the voltage across R_2 will be the same as the voltage across the parallel combination.

$$V_{R_2(1)} = -\left(\frac{0.8\ \text{k}\Omega}{0.8\ \text{k}\Omega + 2.4\ \text{k}\Omega}\right)(16\ \text{V}) = -4.00\ \text{V}$$

- The negative sign in the preceding calculation simply indicates that the voltage across the resistor due to the first **source is opposite to the assumed reference polarity**.

(**b) Consider the current source *I*.**

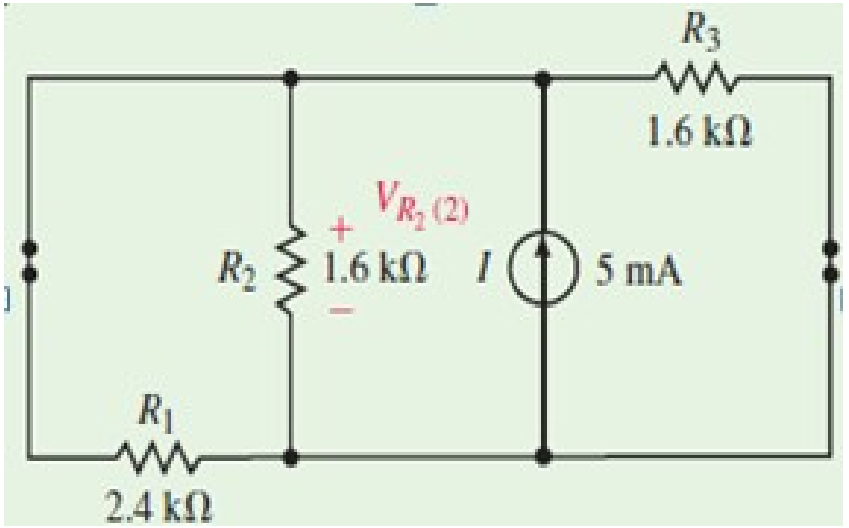
The **total resistance** "seen" by the current source is:

By applying ohm's law

$$R_T = R_1 \| R_2 \| R_3 = 0.6\ \text{k}\Omega$$

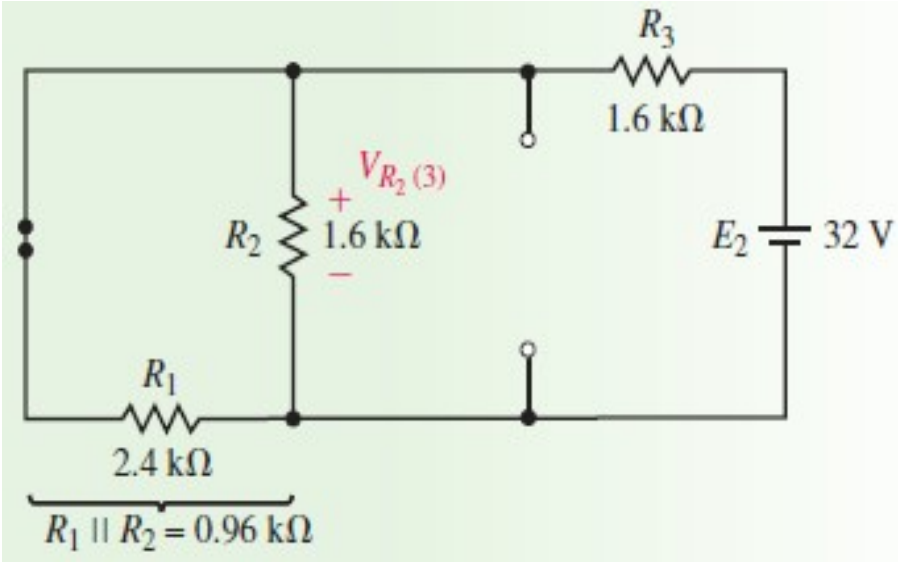
The resulting voltage across R_2 is

$$V_{R_2(2)} = (0.6\ \text{k}\Omega)(5\ \text{mA}) = 3.00\ \text{V}$$



Both have the same voltage as they are all in parallel

(c) Consider the voltage Source *E2* due to the 32 V.



Then, the voltage across R2, is given by VDR as:

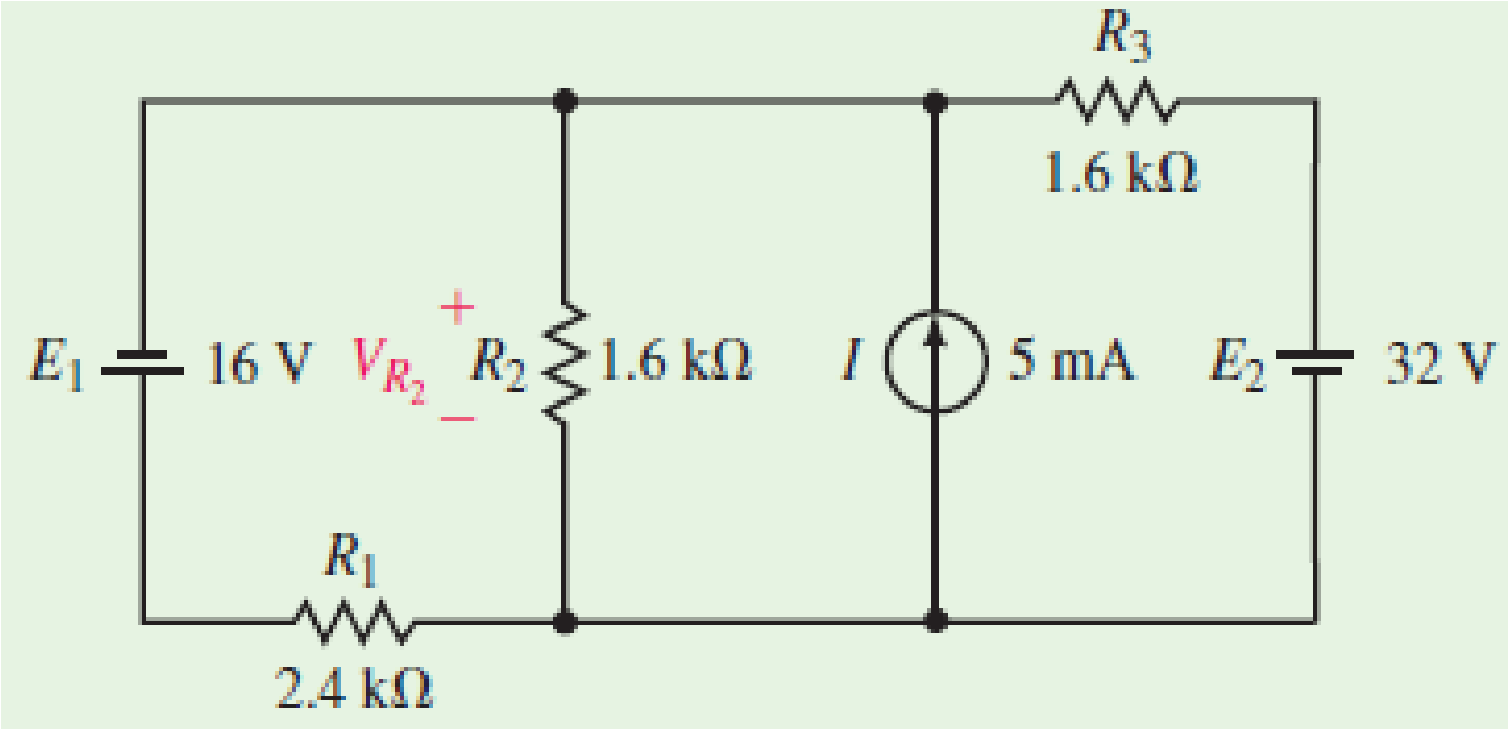
$$V_{R_2(3)} = \left(\frac{0.96\ \text{k}\Omega}{0.96\ \text{k}\Omega + 1.6\ \text{k}\Omega}\right)(32\ \text{V}) = 12.0\ \text{V}$$

By superposition, the resulting voltage is

$$V_{R_2} = -4.0\ \text{V} + 3.0\ \text{V} + 12.0\ \text{V} = 11.0\ \text{V}$$

- **Example for Enhancement**

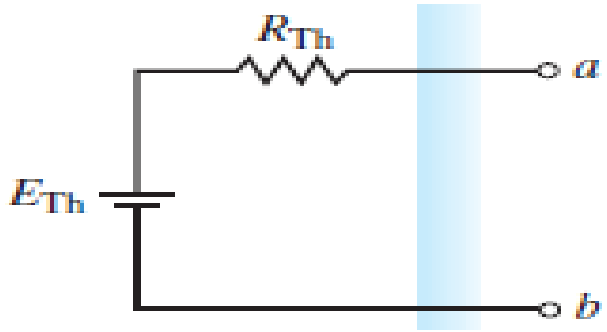
Use the superposition theorem to determine the voltage across $\mathbf{R_1}$ and $\boldsymbol{R_3}$ in the following circuit.



Thévenin's Theorem

- **Thévenin's theorem** allows even the most complicated circuit to be **reduced to a single voltage source and a single resistance**.
- **Thévenin's theorem** is a circuit analysis technique that reduces any linear bilateral network to an equivalent circuit **having only one voltage source and one series resistor.**
- The resulting two-terminal circuit is equivalent to the original circuit when connected to any external branch or component (Load).
- In summary, Thévenin's theorem is simplified as follows:
  - *"Any linear bilateral network may be reduced to a simplified two-terminal circuit consisting of a single voltage source in series with a single resistor".*

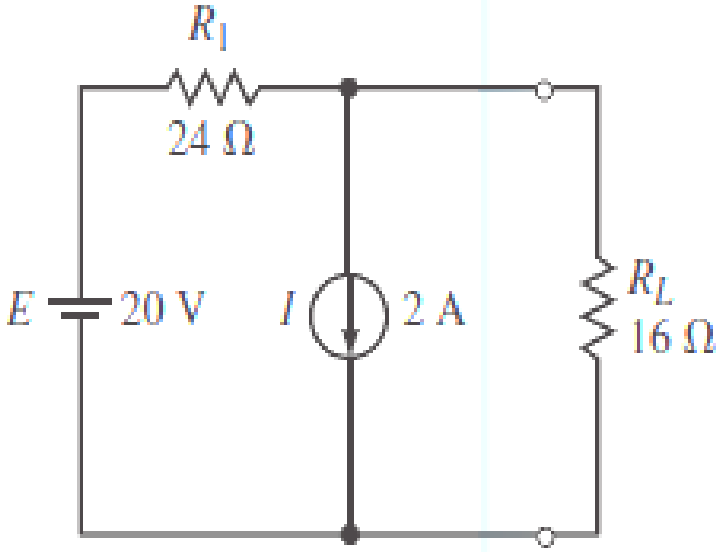
- Reduced Circuit to a single voltage source and a single series resistance



- **Steps to converts any circuit into its Thévenin equ. ckt**

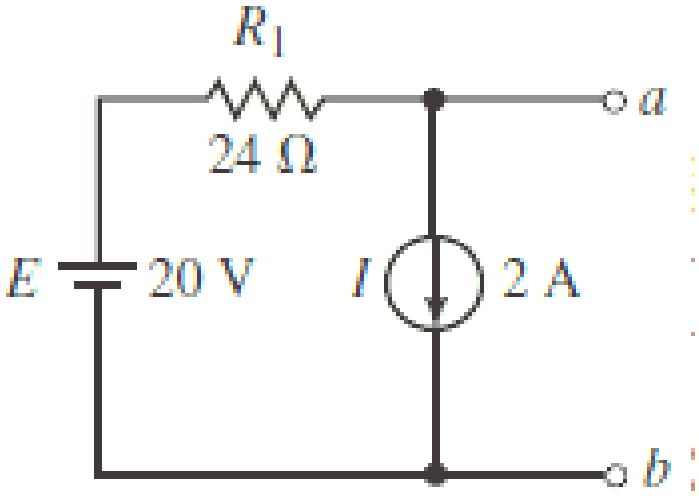
1. Identify and remove the load from the circuit.
2. Label the resulting two terminals (as **a** and **b**).
3. Set all sources in the circuit to zero: Voltage sources are set to zero by replacing them with short circuits, Current sources are set to zero by replacing them with open circuits.
4. Determine the **Thévenin equivalent resistance**, $\mathbf{R_{Th}}$, by calculating the resistance "seen" between terminals **a** and **b**. Redraw the circuit.
5. Replace the sources removed in Step 3, and determine the open-circuit voltage between the terminals. If the circuit has more than one source, it may be necessary to use the superposition theorem. In that case, it will be necessary to determine the open-circuit voltage due to each source separately and then determine the combined effect. The resulting open-circuit voltage will be the value of the **Thévenin voltage**, $\boldsymbol{E_{Th}}$.
6. Draw the Thévenin equivalent circuit using $\boldsymbol{R_{Th}}$ **and** $\boldsymbol{E_{th}}$ and include that portion of the network removed in Step 1.

- **Example (1):**
- Determine the Thévenin equivalent circuit external to the resistor R_L for the circuit. Use the Thévenin equivalent circuit to calculate the current through R_L.

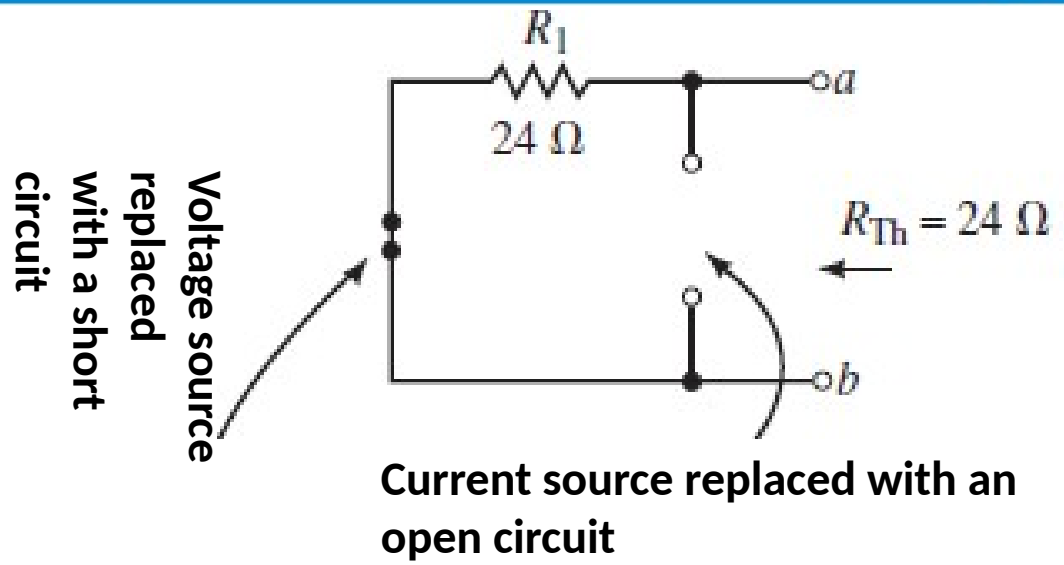


Solution:

Steps 1 and 2: Removing the load resistor from the circuit and labelling the remaining terminals,

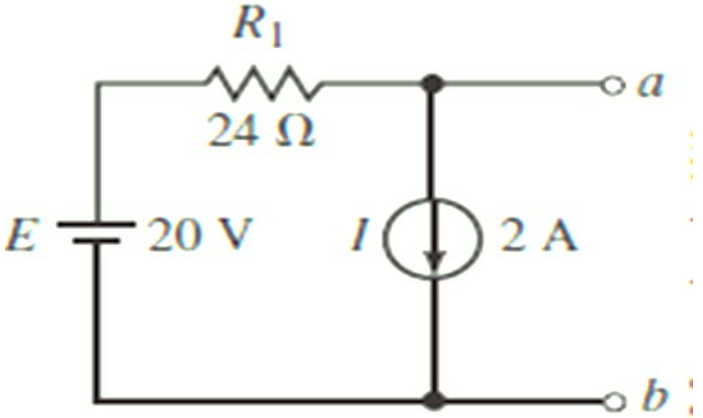


- **Step 3:** Setting the sources to zero

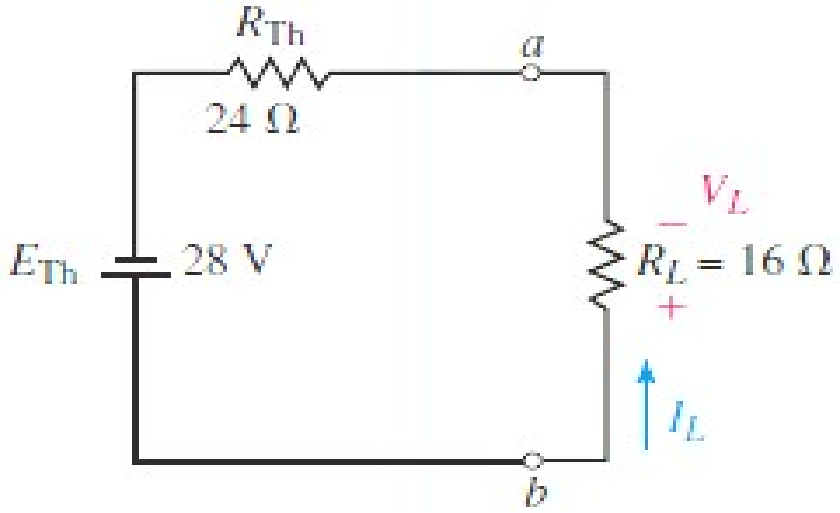


- **Step 4:** The Thévenin resistance between the terminals is R_{TH} =24 .
- **Step 5:** From the figure on the right, the open-circuit voltage between terminals *a* and *b* is:

$$V_{ab} = 20\,\text{V} - (24\,\Omega)(2\,\text{A}) = -28.0\,\text{V}$$



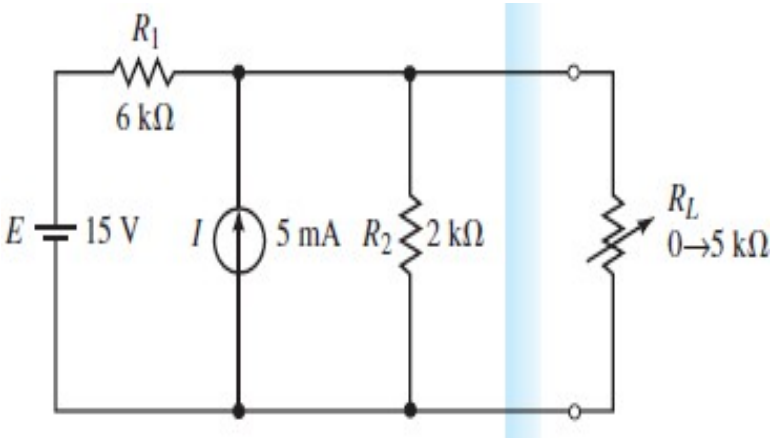
- **Step 6:** The resulting Thévenin equivalent circuit is:



Hence, Using this Thévenin equivalent circuit, we easily find the current through R_L as:

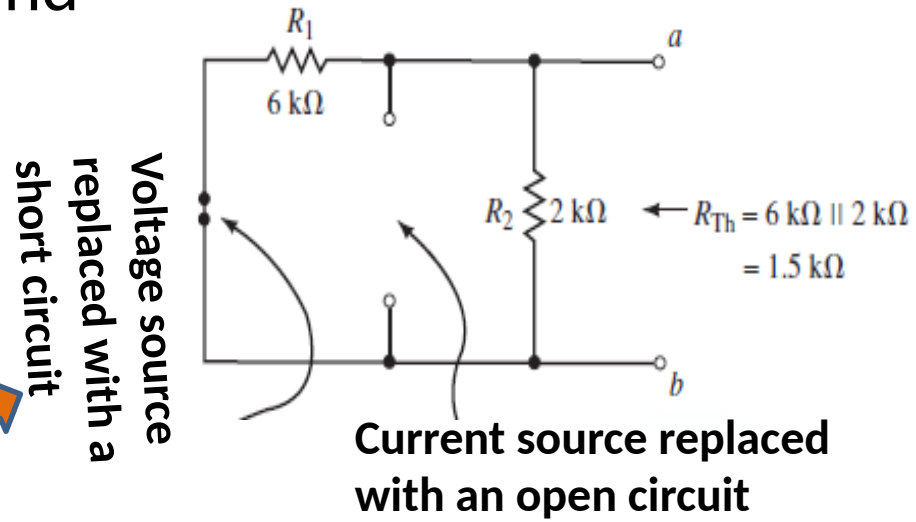
$$I_L = \left(\frac{28\ \text{V}}{24\ \Omega + 16\ \Omega}\right) = 0.700\ \text{A} \quad \text{(upward)}$$

- **Example (2)**
- Find the Thévenin equivalent circuit of the given circuit. Using the equivalent circuit, determine the current through the load resistor when $R_L=0$, $R_L=2\ k$ and $R_L=5\ k$.



Solution

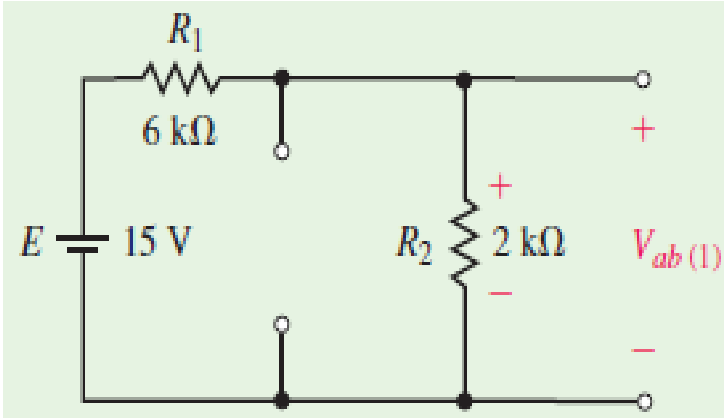
Steps 1, 2, and 3: After removing the load, labelling the terminals, and setting the sources to zero, we have:



Step 4: The Thévenin resistance of the circuit is

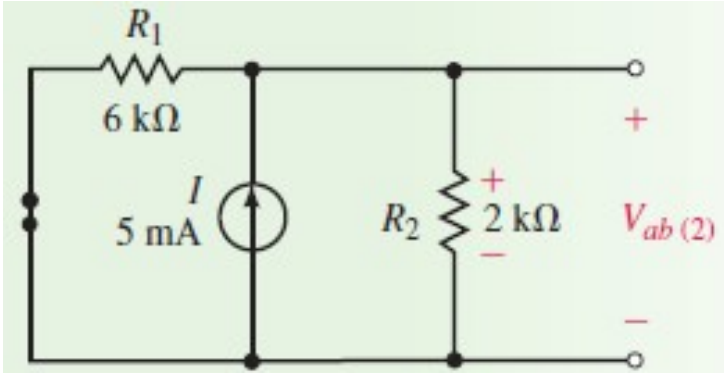
R_{th}= 6k || 2k =1.5k

- **Step 5:** Although several methods are possible, we will use the superposition theorem to find the open-circuit voltage V_{ab} due to the **contribution of the 15-V source (apply VDR).**



$$V_{ab(1)} = \left(\frac{2\,\text{k}\Omega}{2\,\text{k}\Omega + 6\,\text{k}\Omega}\right)(15\text{ V}) = +3.75\text{ V}$$

- Apply superposition theorem to find the open-circuit voltage V_{ab} for determining the **contribution due to the 5-mA source** (apply ohm's law).



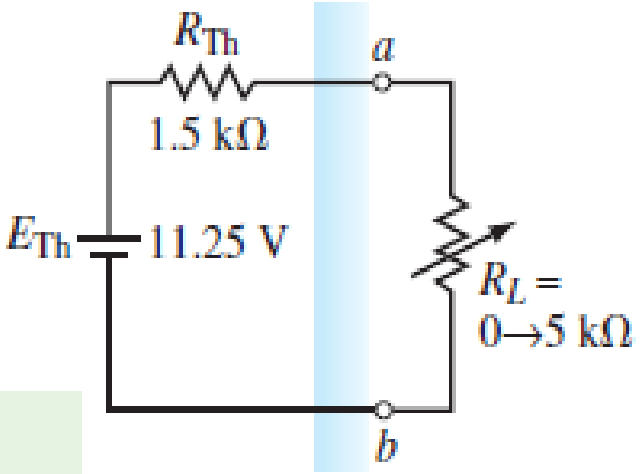
$$V_{ab(2)} = \left(\frac{(2\,\text{k}\Omega)(6\,\text{k}\Omega)}{2\,\text{k}\Omega + 6\,\text{k}\Omega}\right)(5\text{ mA}) = +7.5\text{ V}$$

- Hence,

The Thévenin equivalent voltage is

$$E_{Th} = V_{ab(1)} + V_{ab(2)} = +3.75\ \text{V} + 7.5\ \text{V} = 11.25\ \text{V}$$

- **Step 6:** The resulting Thévenin equivalent circuit is:



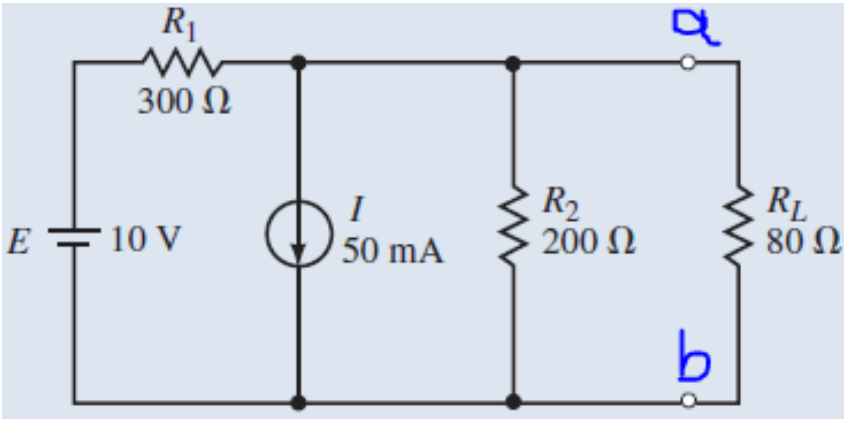
- From the left circuit, it is now an easy matter to determine the current for any value of load resistor as follows **(Apply ohm's law):**

$R_L = 0\ \Omega$: $\quad I_L = \dfrac{11.25\ \text{V}}{1.5\ \text{k}\Omega} = 7.5\ \text{mA}$

$R_L = 2\ \text{k}\Omega$: $\quad I_L = \dfrac{11.25\ \text{V}}{1.5\ \text{k}\Omega + 2\ \text{k}\Omega} = 3.21\ \text{mA}$

$R_L = 5\ \text{k}\Omega$: $\quad I_L = \dfrac{11.25\ \text{V}}{1.5\ \text{k}\Omega + 5\ \text{k}\Omega} = 1.73\ \text{mA}$

(2) Use Thévenin's theorem to determine the current through load resistor R_L for the Circuit of **Fig** below



Norton's Theorem

- Norton's theorem is a circuit analysis technique that is similar to Thévenin's theorem.
- By using this theorem, we reduce **the circuit to a single current source and one parallel resistor**.
- As with the Thévenin equivalent circuit, **the resulting two-terminal circuit is equivalent to the original circuit when connected to any external branch** or component.
- In summary, Norton's theorem may be simplified as **follows:**
  - ***"Any linear bilateral network may be reduced to a simplified two-terminal circuit consisting of a single current source and a single shunt resistor".***

- **Steps to convert any circuit into its Norton equivalent:**

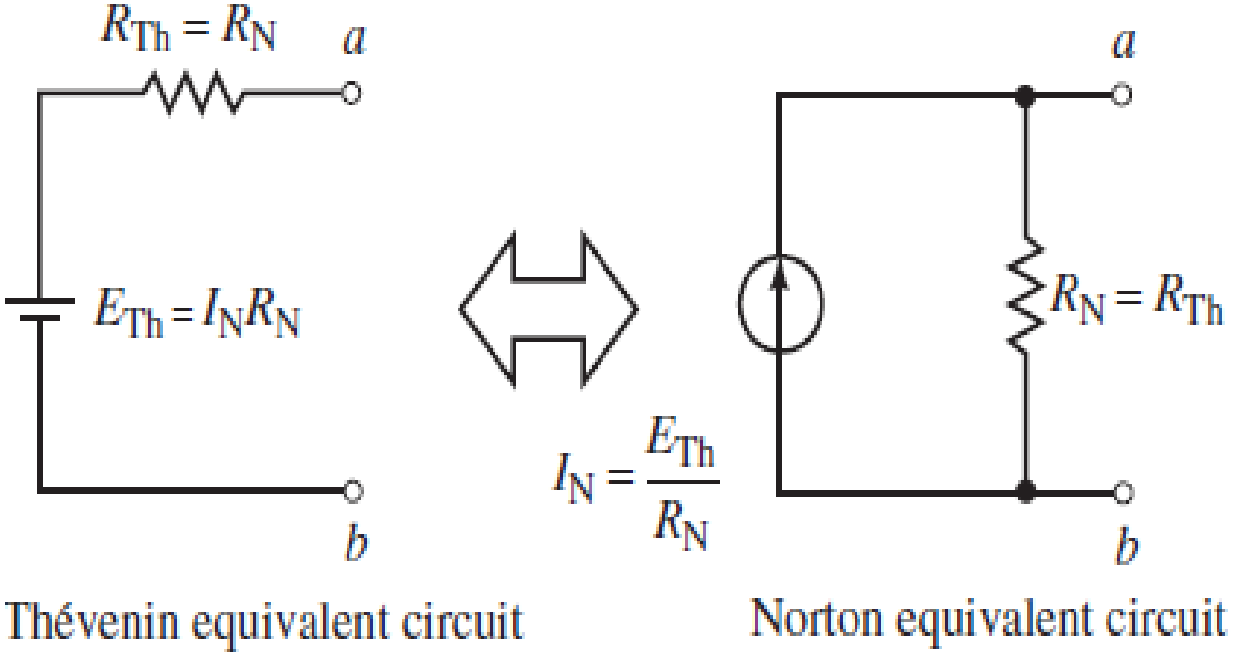
1. Identify and **remove the load** from the circuit
2. Label the resulting two terminals (as **a** and **b**).
3. **Set all sources to zero**. As before, voltage sources are replaced by short circuits, and current sources by open circuits.
4. Determine the Norton equivalent resistance, R_N, by calculating the resistance seen between terminals a and b. Redraw the circuit.
5. Replace the sources removed in Step 3, and determine the current that would occur in a short if the short were connected between terminals **a** and **b.** If **the original circuit has more than one source, it may be necessary to use the superposition theorem**. The resulting short-circuit current will be the value of the Norton current I_N.
6. Sketch the Norton equivalent circuit using the resistance R_N and I_N. Then, include that portion of the network removed in Step 1 (load).

- Also , the Norton equivalent circuit may also be determined directly from the Thévenin equivalent circuit by using the source conversion technique (Source transformation)

- **From the left Circuit, We can calculate the following:**

$$E_{Th} = I_N R_N$$

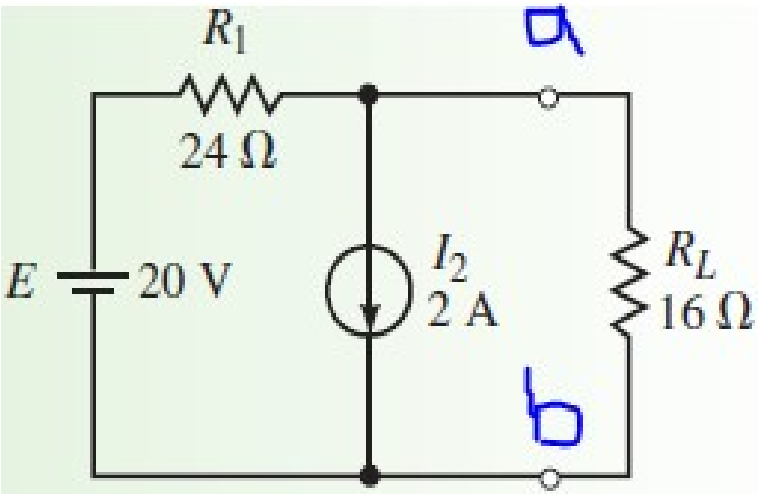
$$I_N = \frac{E_{Th}}{R_{Th}}$$



Thévenin equivalent circuit

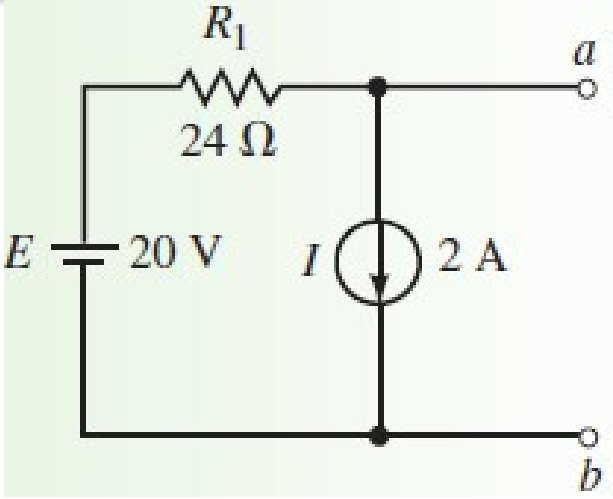
Norton equivalent circuit

- **Example (1):**
- Determine the Norton equivalent circuit external to the resistor R_L for the given circuit. Use the Norton equivalent circuit to calculate the current through R_L. Compare the results to those obtained using Thévenin's theorem.

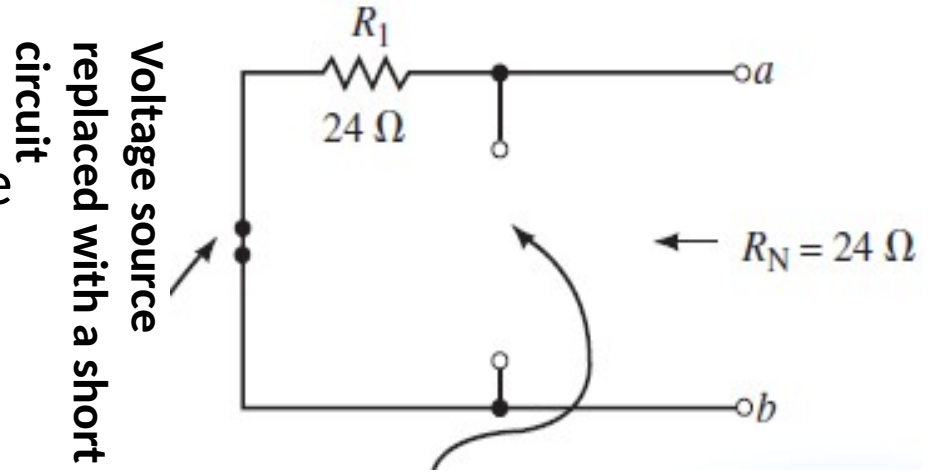


Solution:

- **Steps 1 and 2:** Remove load resistor R_L from the circuit and label the remaining terminals as *a* and *b*.



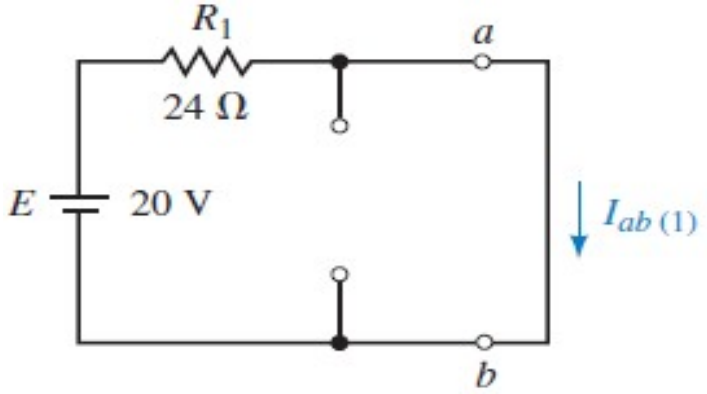
- **Step 3:** Zero the voltage and current sources as shown in the circuit on the right.



Current source replaced with an open circuit

- **Step 4**: The resulting Norton resistance between the terminals is $R_N = R_{ab} = 24$

- **Step 5:** The short-circuit current is determined by first calculating the current through the short due to each source.



(a) Voltage source

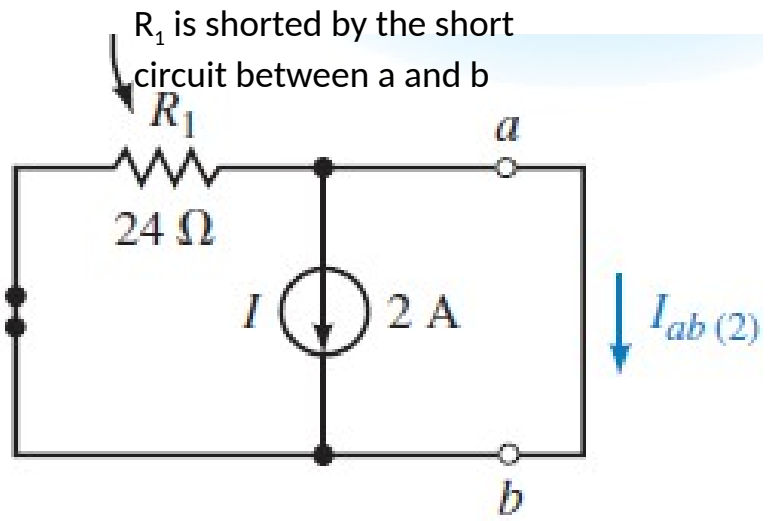
- ***Voltage Source, E***: The current in the short between terminals a and b

$$I_{ab(1)} = \frac{20\ \text{V}}{24\ \Omega} = 0.833\ \text{A}$$

- ***Current Source, I***: By examining the circuit for the current source, we see that the short circuit between terminals **a** and **b** effectively removes $\mathbf{R_1}$ from the circuit.

- Therefore, the current through the short will be $I_{ab(2)} = -2.00\ A$

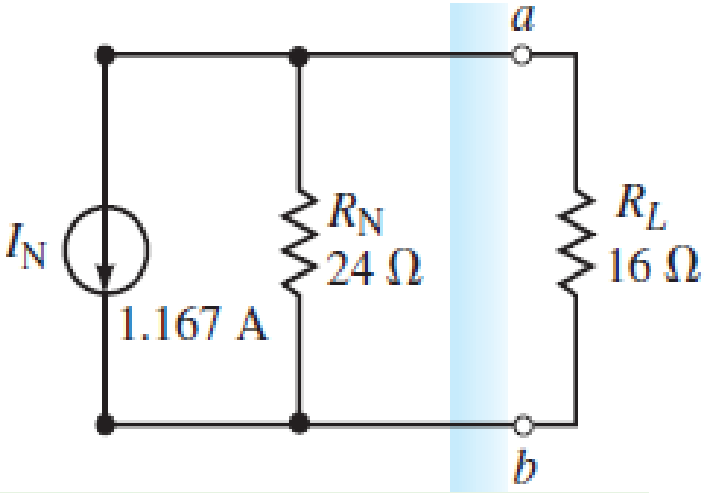
Notice that $\mathbf{R_1}$ is shorted by the short circuit **between a and b**



(b) The current source

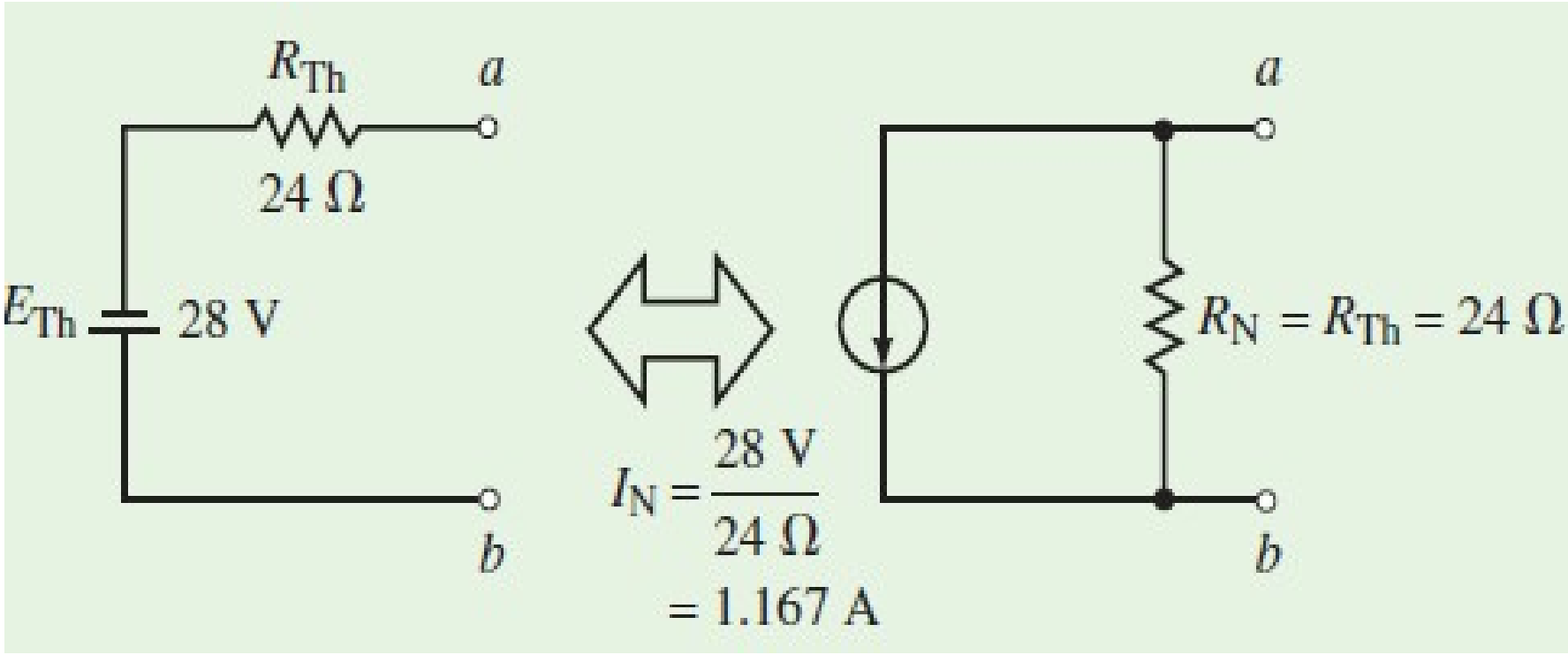
- Now, applying the ***superposition theorem***, we find the Norton current as:
- $I_N = I_{ab(1)} + I_{ab(2)} = 0.833\ A - 2.0A = -1.167\ A$
- As before, the **negative sign** indicates that the short-circuit current is actually from terminal ***b*** toward terminal *a*.

- **Step 6:** The resultant Norton equivalent circuit is shown.
- Now we can easily find the current through load resistor $\mathbf{R_L}$ by using the current divider rule:

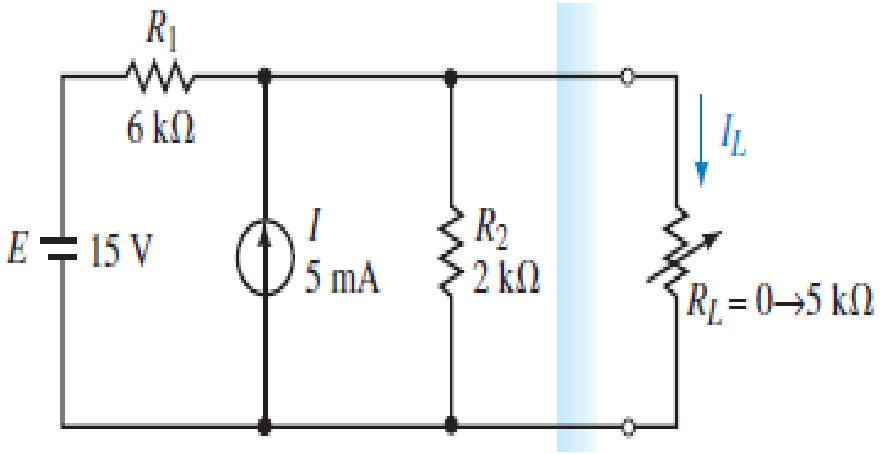


$$I_L = \left(\frac{24\ \Omega}{24\ \Omega + 16\ \Omega}\right)(1.167\ \text{A}) = 0.700\ \text{A} \quad \text{(upward)}$$

- We see that the same result was obtained by using the Thévenin equivalent circuit (**See Source transformation**).

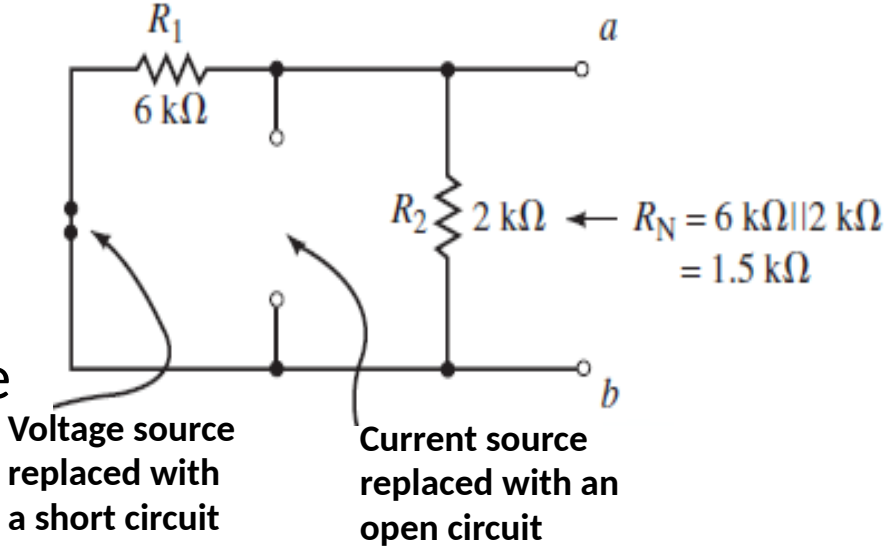


- **Example (2):**
- Find the Norton equivalent of the circuit external to resistor R_L in the circuit. Use the equivalent circuit to determine the load current I_L when R_L=0, 2 k , and 5 k .

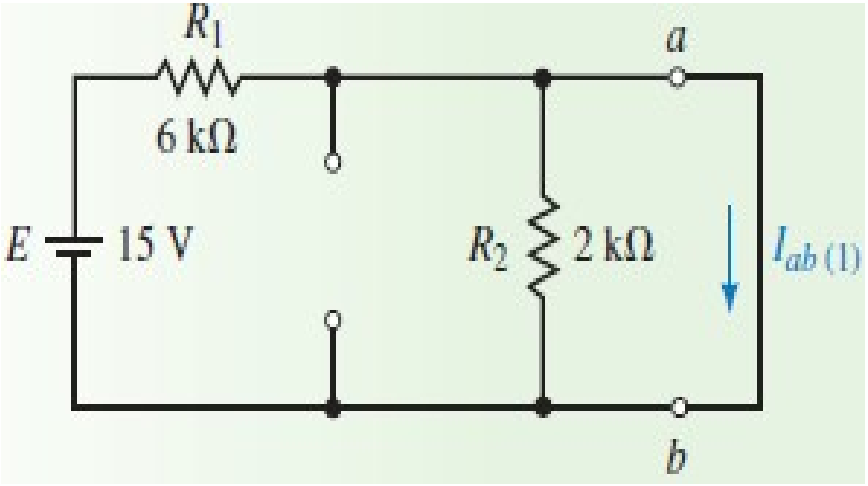


Solution

Steps 1, 2, and 3: After removing the load resistor, labelling the remaining two terminals **a** and **b**, and setting the sources to zero, we have the following circuit

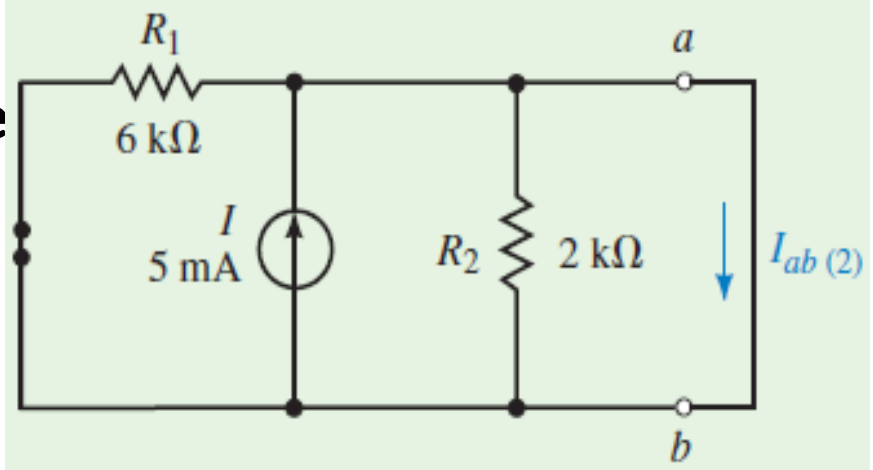


- **Step 4**: The Norton resistance of the circuit is found as $R_N = 6\ k\Omega \,||\, 2\ k\Omega = 1.5\ k\Omega$.
- **Step 5:** The value of the Norton constant-current source is found by determining the current effects due to each independent source acting on a short circuit between terminals ***a*** and ***b***.
- ***Voltage Source, E***: A short circuit between terminals **a** and **b** **eliminates resistor R_2** from the circuit. The short-circuit **current** due to the voltage source is:



$$I_{ab(1)} = \frac{15\ \text{V}}{6\ \text{k}\Omega} = 2.50\ \text{mA}$$

***Current Source, I*:** The short circuit between terminals **a** and **b eliminate both resistors R_1 and R_2**. The short-circuit current due to the current source is therefore:

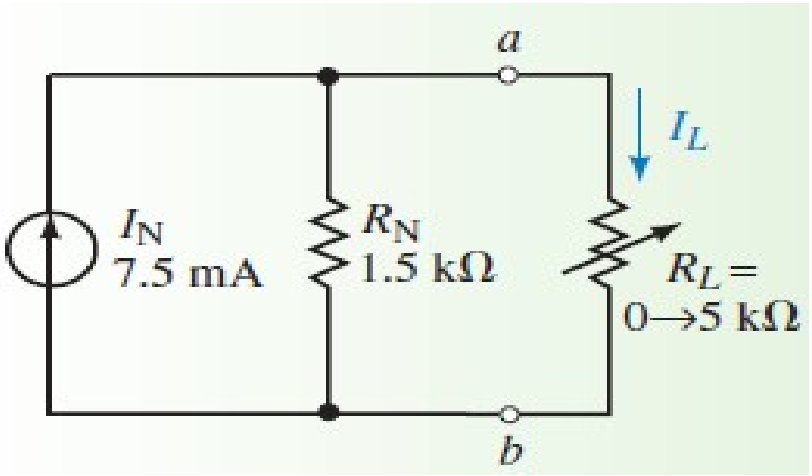


$$I_{ab(2)} = 5.00\ \text{mA}$$

- The resultant I_N is found from superposition as:

$$I_N = I_{ab(1)} + I_{ab(2)} = 2.50\ \text{mA} + 5.00\ \text{mA} = 7.50\ \text{mA}$$

- **Step 6**: The Norton equivalent circuit is:



- **Using CDR, we have:**

Let $R_L = 0$: The current I_L must equal the source current, and so

$$I_L = 7.50 \text{ mA}$$

Let $R_L = 2\ \text{k}\Omega$: The current I_L is found from the current divider rule as

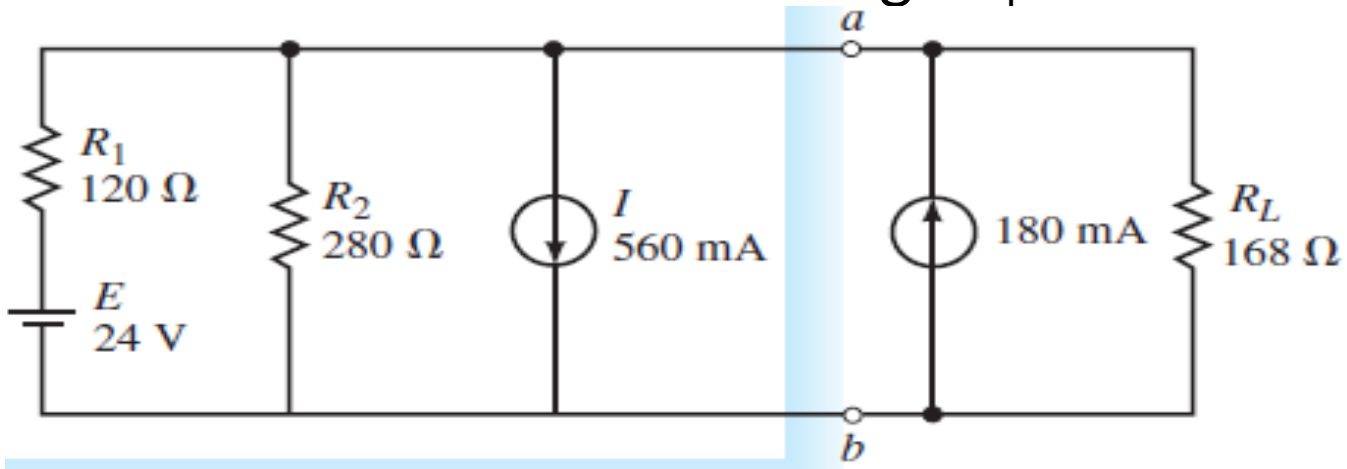
$$I_L = \left(\frac{1.5\ \text{k}\Omega}{1.5\ \text{k}\Omega + 2\ \text{k}\Omega}\right)(7.50\ \text{mA}) = 3.21\ \text{mA}$$

Let $R_L = 5\ \text{k}\Omega$: Using the current divider rule again, the current I_L is found as

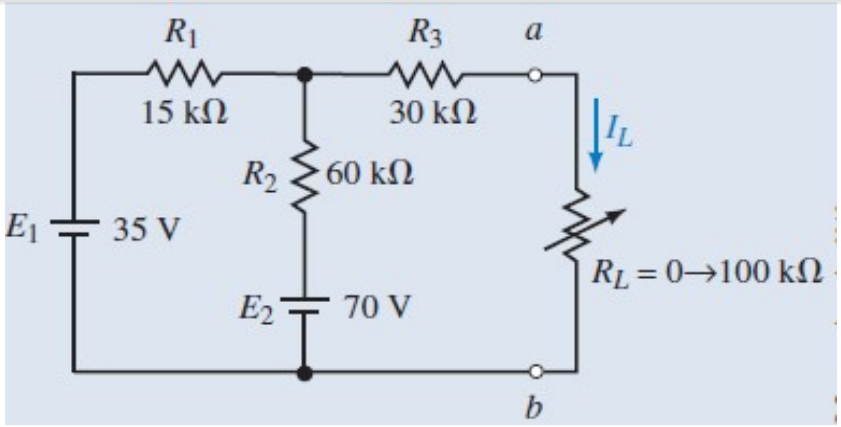
$$I_L = \left(\frac{1.5\ \text{k}\Omega}{1.5\ \text{k}\Omega + 5\ \text{k}\Omega}\right)(7.50\ \text{mA}) = 1.73\ \text{mA}$$

- **Example for Enhancement:**

1. Find the Norton equivalent circuit external to terminals **a** and **b**. Determine the current through R_l.



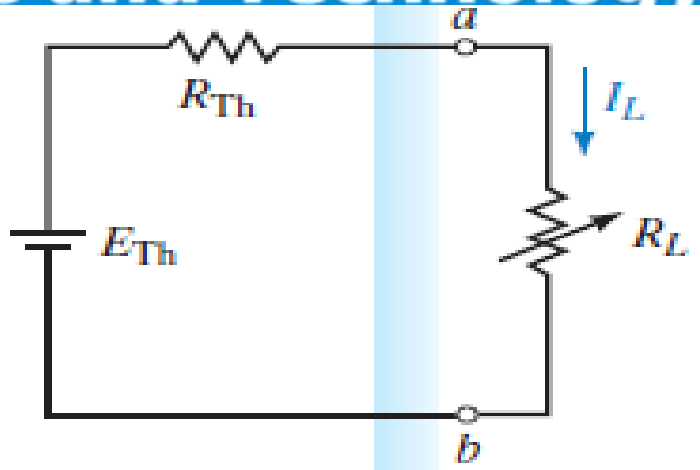
2. Find the Norton equivalent external to R_L in the circuit on the left. Solve for the current I_L when R_L=0, 10 k , 50 k , and 100 k .



Maximum Power Transfer Theorem

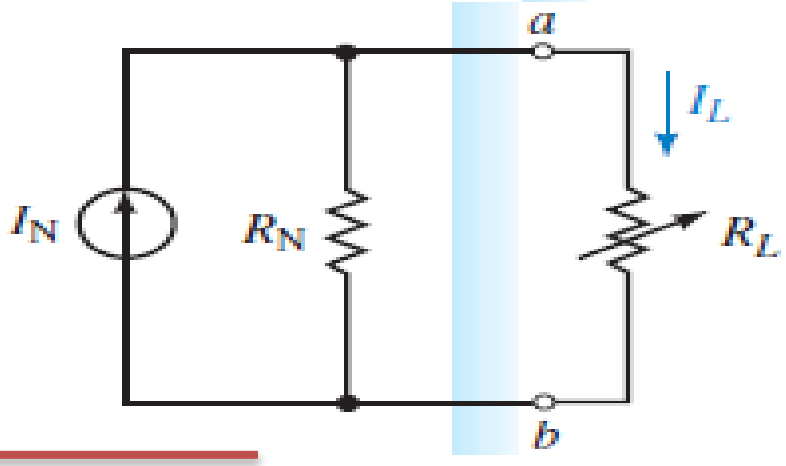
- In amplifiers and in most communication circuits such as radio receivers and transmitters, it is often desired that the load receive the maximum amount of power from a source.
- The maximum power transfer theorem states the following:
    - *"A load resistance will receive maximum power from a circuit when the resistance of the load is exactly the same as the **Thévenin (Norton)** resistance looking back at the circuit".*
- The maximum power will **occur when** $R_L = R_{Th} = R_N$

- By examining the Thevenin's equivalent circuits and by using VDR, the power delivered to the load is given by:
- V drop across R_L (use VDR)



$$P_L = \frac{V^2}{R_L} = \frac{\left(\frac{R_L}{R_L+R_{TH}} \times E_{TH}\right)^2}{R_L}$$

$$\boldsymbol{P_L = \frac{V^2}{R_L} = \frac{R_L \times (E_{TH})^2}{(R_L+R_{TH})^2}}$$



Then By Considering **The Norton's Theorem, and by applying CDR**, the Power delivered to the load is Given by :

$$P_L = I^2 R_L$$

$$= \left(\left(\frac{R_N I_N}{R_L + RN} \right)^2 \times \boldsymbol{R_L} \right)$$

- Under **maximum power conditions ($R_L=R_{Th}=R_N$),** the preceding equations may be used to determine the maximum power delivered to the load as:

$$P_{max} = \frac{E_{Th}^2}{4R_{Th}}$$

By Thevenin's Theorem

$$P_{max} = \frac{I_N^2 R_N}{4}$$

By Norton's Theorem

Example: Consider the Circuit of:

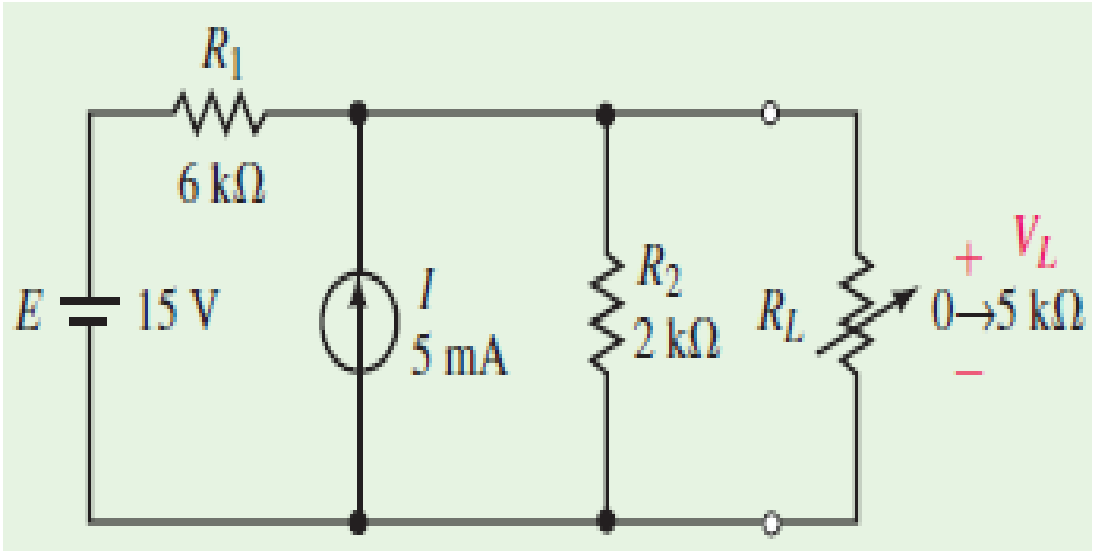
a. Determine the value of load resistance required to ensure that maximum power is transferred to the load.

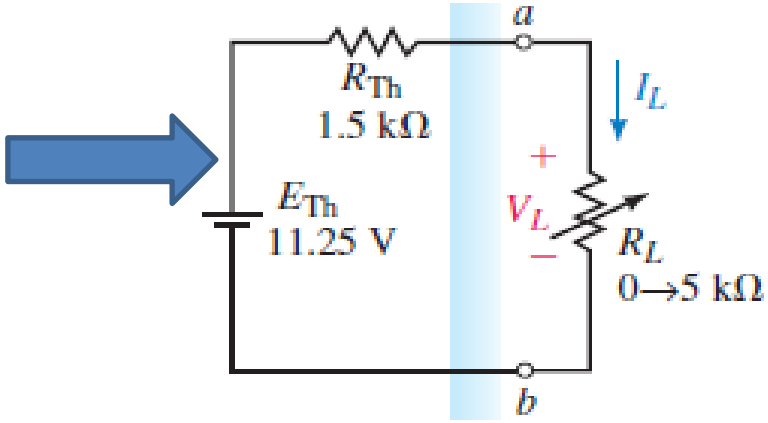
b. Find V_L, I_L , and P_L when maximum power is delivered to the load.

Example (1): Consider the following Circuit:

a) Determine the value of load resistance required to ensure that maximum power is transferred to the load.

b) Find V_L, I_L, and P_L when maximum power is delivered to the load.





The equivalent circuit external to the load.

Refer to slide 17-19

(a) Maximum power will be transferred to the load when **$R_{TH}=R_L=1.5$ k .**

(b) At maximum power and by letting **$R_L=1.5$ k ,** the voltage across the Load (V_L)is divided by 2, i.e, ½V_L is dissipated in R_L & another 1/2V_L is dissipated in R_{TH}

$$V_L = \frac{E_{Th}}{2} = \frac{11.25\ \text{V}}{2} = 5.625\ \text{V}$$

$$I_L = \frac{5.625\ \text{V}}{1.5\ \text{k}\Omega} = 3.750\ \text{mA}$$

The power delivered to the load is found as

$$P_L = \frac{V_L^2}{R_L} = \frac{(5.625\ \text{V})^2}{1.5\ \text{k}\Omega} = 21.1\ \text{mW}$$

Or, alternatively using current, we calculate the power as

$$P_L = I_L^2 R_L = (3.75\ \text{mA})^2(1.5\ \text{k}\Omega) = 21.1\ \text{mW}$$

In solving this problem, we could just as easily have used the Norton equivalent circuit to determine required values.

Efficiency

- **The efficiency ( )** is defined as the ratio of output power to input power:

$$\eta = \frac{P_{out}}{P_{in}} \quad \textbf{or} \quad \eta = \frac{P_{out}}{P_{in}} \times 100\% \quad \textbf{:As percentages}$$

- By using the maximum power transfer theorem, we see that under the ***condition of maximum power*** the **efficiency ( )** of the circuit is:

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

$$= \frac{\frac{E_{Th}^2}{4R_{Th}}}{\frac{E_{Th}^2}{2R_{Th}}} \times 100\% = 0.500 \times 100\% = 50\%$$

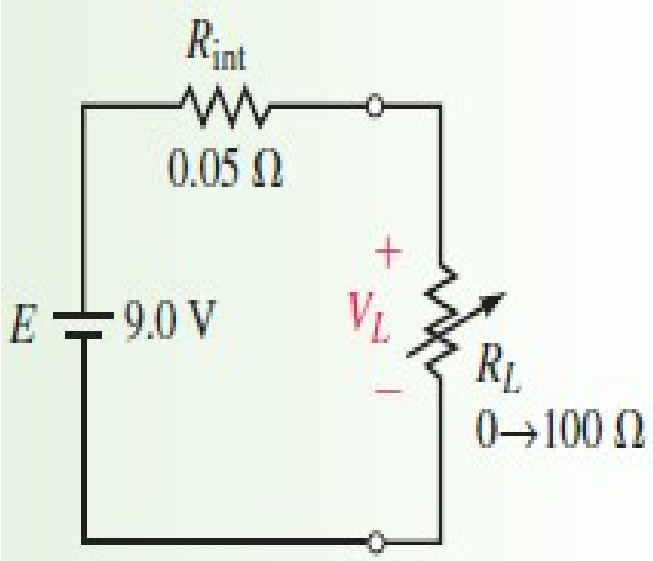
Note: $R_{TH}=R_L$, so in the Equi. ckt: $R_{TH}+R_{TH}=2R_{TH}$

Example (2): For the following the circuit:

a) Determine the value of R_L needed for maximum power transfer.

b) Determine terminal voltage V_L and the efficiency when the value of the load resistor is R_L=50 .

c) Determine terminal voltage V_L and the efficiency when the value of the load resistor is R_L=100



$$\eta = (V_L^2/R_L)/(E_s^2/(R_L+R_{TH}))$$

Solution

a. For maximum power transfer, the load resistor will be given as $R_L = 0.05\ \Omega$. At this value of load resistance, the efficiency will be only 50%.

b. For $R_L = 50\ \Omega$, the voltage appearing across the output terminals of the voltage source is

$$V_L = \left(\frac{50\ \Omega}{50\ \Omega + 0.05\ \Omega}\right)(9.0\ \text{V}) = 8.99\ \text{V}$$

The efficiency is

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

$$= \frac{\frac{(8.99\ \text{V})^2}{50\ \Omega}}{\frac{(9.0\ \text{V})^2}{50.05\ \Omega}} \times 100\%$$

$$= \frac{1.6168\ \text{W}}{1.6184\ \text{W}} \times 100\% = 99.90\%$$

Notes:

- To have a high efficiency, the ***load resistance should be much larger*** than the resistance of the source (typically $R_L \geq 10R_{int}$).
- To *ensure maximum power transfer*, then the load resistance should equal the source resistance ($R_L = R_{int}$).

c. For R_L=100 , the voltage appearing across the output terminals of the voltage source

$$V_L = \left(\frac{100\ \Omega}{100\ \Omega + 0.05\ \Omega}\right)(9.0\ \text{V}) = 8.995\ 50\ \text{V}$$

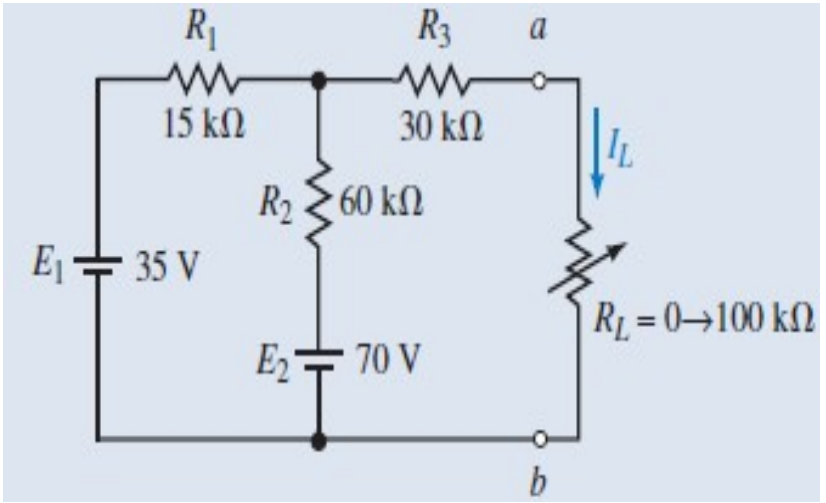
The efficiency is

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

$$= \frac{\frac{(8.9955\ \text{V})^2}{100\ \Omega}}{\frac{(9.0\ \text{V})^2}{100.05\ \Omega}} \times 100\%$$

$$= \frac{1.6168\ \text{W}}{1.6184\ \text{W}} \times 100\% = 99.95\%$$

Example for Enhancement:



(1) Refer to the circuit on the right side.

a) For what value of R_L will the load receive maximum power?

b) Determine the power when $R_L=R_N$, when $R_L=25$ k , and when $R_L=50$ k .

---

(2) A Thévenin equivalent circuit consists of $E_{Th}=10$ V and $R_{Th}=2$k . Determine the efficiency of the circuit when

a. $R_L=R_{Th}$

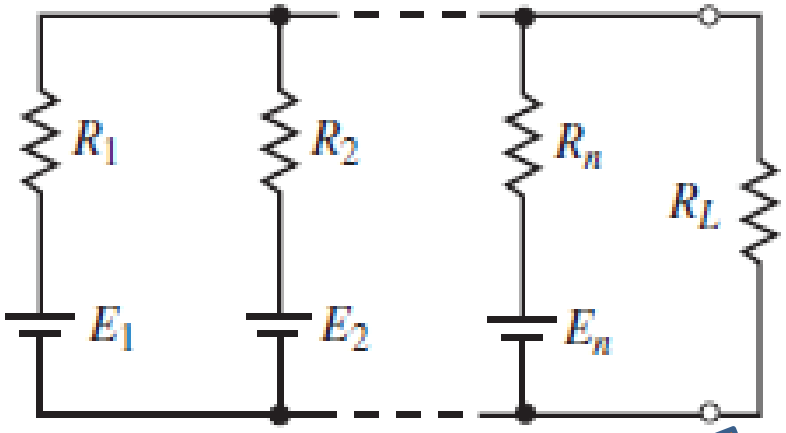
b. $R_L=0.5R_{Th}$

c. $R_L=2R_{Th}$

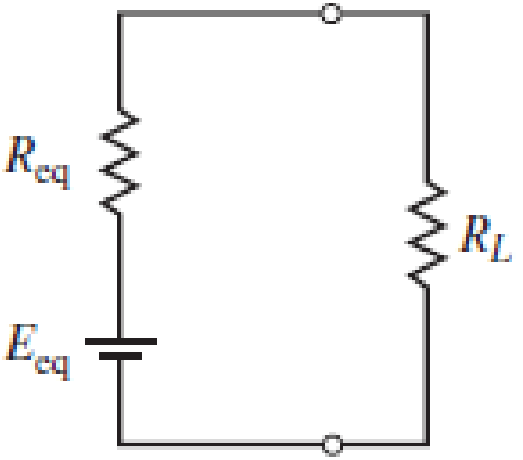
Millman's Theorem

- **Millman's theorem** is used to simplify circuits having several parallel voltage sources .
- Although any of the other theorems developed in this chapter will work in this case, **Millman's theorem provides a much simpler and more direct equivalent.**
- The voltage sources may be replaced with a single equivalent source (**Series of E_{eq}, R_{eq} and R_L**).

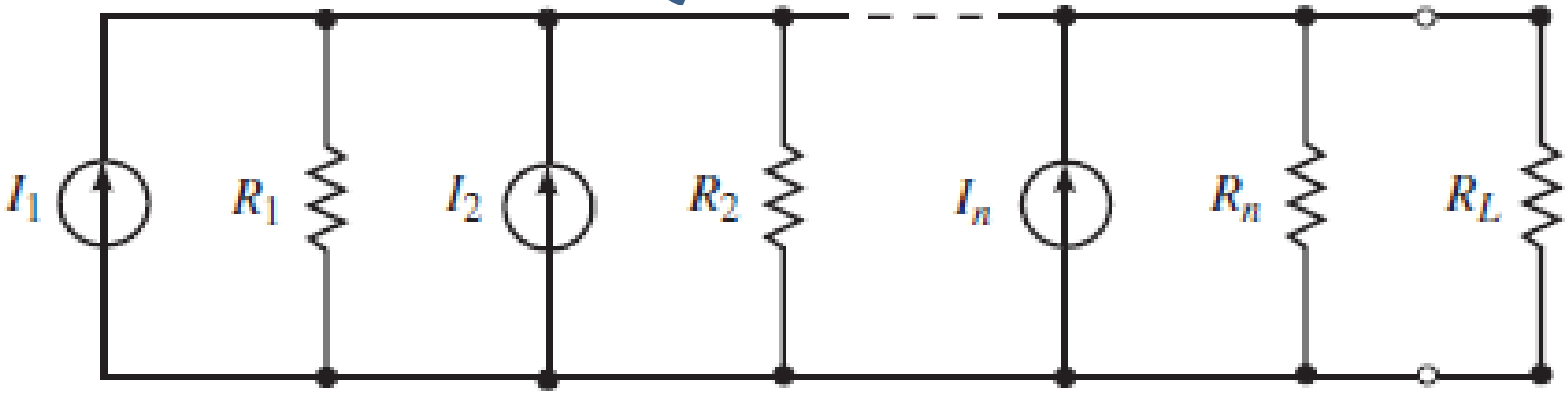
- To find the values of the **equivalent voltage source E_{eq} and series resistance R_{eq},** we need to ***convert*** each of the **voltage sources** into their equivalent **current sources**.
- The value of each current source would be determined by using Ohm's law (**i.e., $I_1=E_1/R_1$, $I_2=E_2/R_2$, etc.**)



Circuits having several parallel voltage sources



Single equivalent source (Series of E_{eq}, R_{eq} and R_L)



Converted voltage sources into their equivalent current sources

- It is now possible to **replace the *n* current sources** with a single current source having a magnitude given as:

$$I_{eq} = \sum_{x=0}^{n} I_x = I_1 + I_2 + I_3 + \cdots + I_n$$

- Write it as:

$$I_{eq} = \frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} + \cdots + \frac{E_n}{R_n}$$

- After Converting & Summing up currents, the E_{eq} is written as follows:

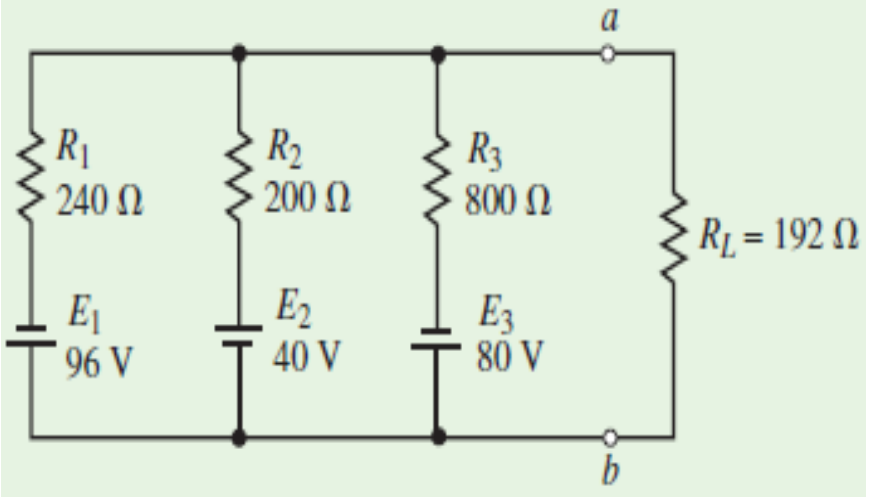
$$R_{eq} = \frac{1}{G_{eq}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}}$$

- The general expression for the equivalent voltage is:

$$E_{eq} = I_{eq}R_{eq} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} + \cdots + \frac{E_n}{R_n}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}}$$

Example:

- Use Millman's theorem to simplify the circuit so that it has only a single source.
- Use the simplified circuit to find the current in the load resistor, R_L.



- **Solution:** We express the equivalent voltage source as:

$$V_{ab} = E_{eq} = \frac{\frac{-96\text{ V}}{240\ \Omega} + \frac{40\text{ V}}{200\ \Omega} + \frac{-80\text{ V}}{800\ \Omega}}{\frac{1}{240\ \Omega} + \frac{1}{200\ \Omega} + \frac{1}{800\ \Omega}}$$

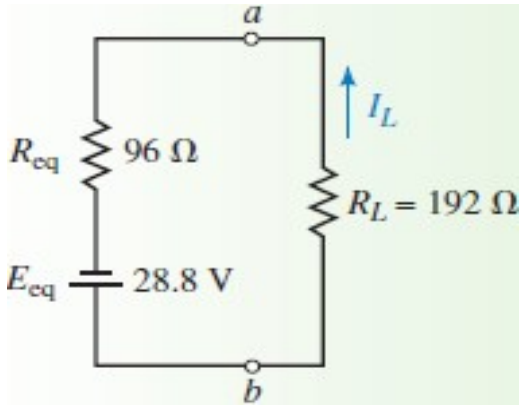
$$V_{ab} = \frac{-0.300}{10.42\text{ mS}} = -28.8\text{ V}$$

- The equivalent resistance is

$$R_{eq} = \frac{1}{\frac{1}{240\ \Omega} + \frac{1}{200\ \Omega} + \frac{1}{800\ \Omega}} = \frac{1}{10.42\text{ mS}} = 96\ \Omega$$

- Current through the load resistor:

$$I_L = \frac{28.8\text{ V}}{96\ \Omega + 192\ \Omega} = 0.100\text{ A} = 100\text{ mA} \quad \text{(upward)}$$



Equivalent circuit

Reciprocity Theorem

- The reciprocity theorem is a theorem that can only be ***used with single-source circuits***. This theorem, however, **may be applied to either voltage sources or current sources**. The theorem states the following:
- **Voltage Sources:**
  - *A voltage source causing a current **I** in any branch of a circuit may be removed from the original location and placed into that branch having the current **I**. The voltage source in the new location will produce a current in the original source location that is exactly equal to the originally calculated current, **I**.*
- When applying the reciprocity theorem for a voltage source, the following steps must be followed:
  1) The **voltage source** is replaced by a short circuit in the original location.
  2) **The polarity of the source** in the new location is such that the current direction in that branch remains unchanged.

- **Current Sources:**
  - ***"A current source causing a voltage V at any node of a circuit may be removed from the original location and connected to that node. The current source in the new location will produce a voltage in the original source location that is exactly equal to the originally calculated voltage, V".***
- When applying the reciprocity theorem for a current source, the following conditions must be met:

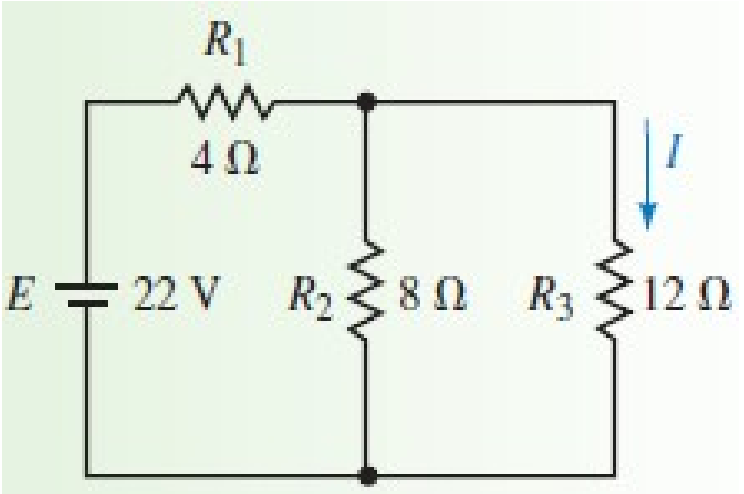
1) **The current source** is replaced by an open circuit in the original location.
2) The direction of the source in the new location is such that the polarity of the voltage at the node to which the current source is now connected remains unchanged.

- **Example (1):**

Consider the following circuit:

(a) Calculate the current ***I***.

(b) Remove voltage source ***E*** and place it into the branch with $\mathbf{R_3}$. Show that the current through the branch which formerly had ***E*** is now the same as the current ***I***.



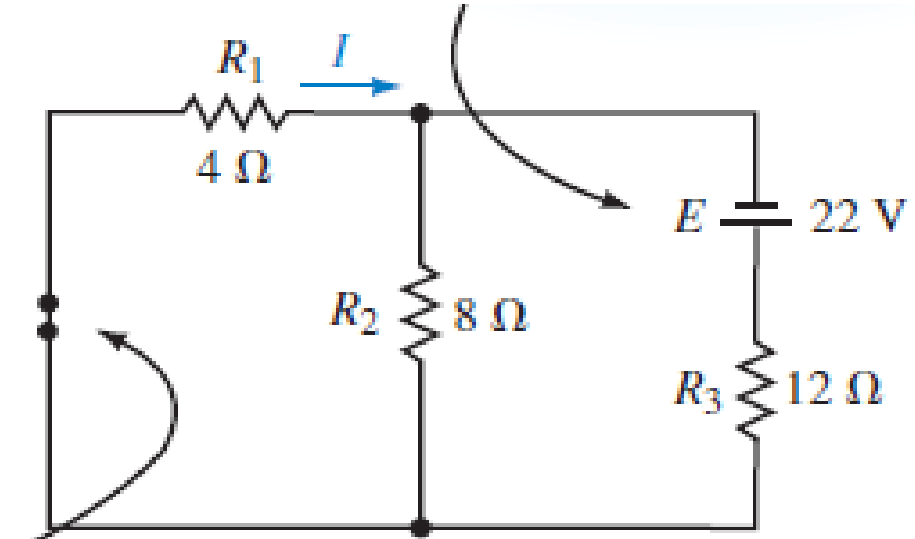
Solution

$$\text{a. } V_{12\,\Omega} = \left(\frac{8\,\Omega \| 12\,\Omega}{4\,\Omega + (8\,\Omega \| 12\,\Omega)}\right)(22\text{ V}) = \left(\frac{4.8}{8.8}\right)(22\text{ V}) = 12.0\text{ V}$$

$$I = \frac{V_{12\,\Omega}}{12\,\Omega} = \frac{12.0\text{ V}}{12\,\Omega} = 1.00\text{ A}$$

- **Now removing the voltage source** from its original location and moving it into the branch containing the current *I*, we obtain the circuit:

Polarity of the source is such that the current direction remains unchanged.



When E is removed it is replaced by a short circuit.

- We determine *the current I* as follows:

$$V_{4\,\Omega} = \left(\frac{4\ \Omega \| 8\ \Omega}{12\ \Omega + (4\ \Omega \| 8\ \Omega)}\right)(22\ \text{V}) = \left(\frac{2.\overline{6}}{14.6}\right)(22\ \text{V}) = 4.00\ \text{V}$$

$$I = \frac{V_{4\,\Omega}}{4\ \Omega} = \frac{4.00\ \text{V}}{4\ \Omega} = 1.00\ \text{A}$$

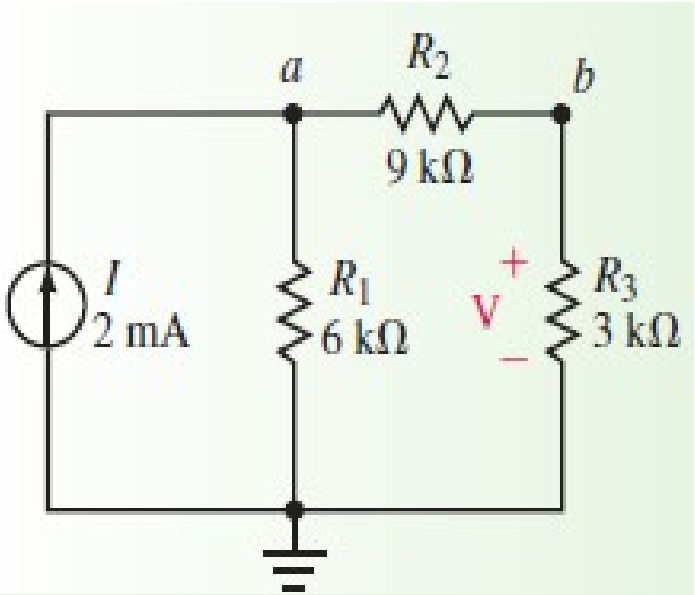
- **We see that the reciprocity theorem does indeed apply!!!**

- **Example (2):**

Consider the circuit shown.

(a) Determine the ***voltage V*** across resistor $\mathbf{R_3}$.

(b) Remove the current ***source I*** and place it between node b and the reference node. Show that the voltage across the former location of the current source (node a) is now the same as the voltage V.



Solution

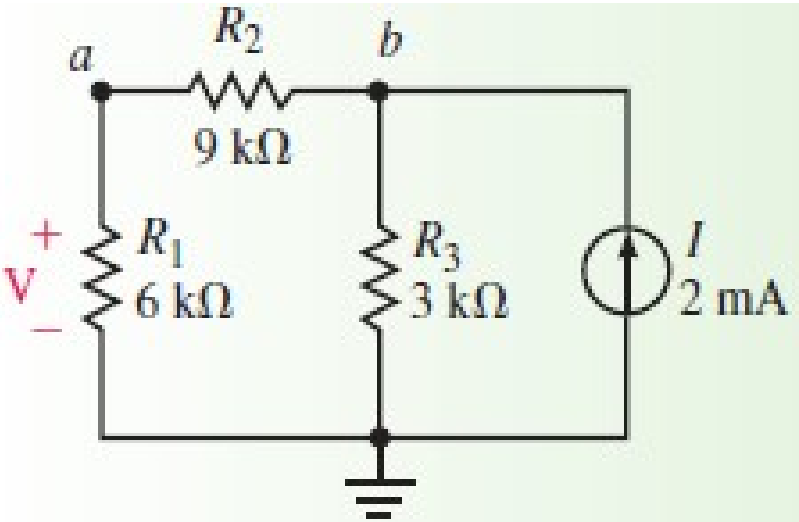
a. The node voltages for the circuit [redacted]re determined as follows:

$$R_T = 6\ \text{k}\Omega \| (9\ \text{k}\Omega + 3\ \text{k}\Omega) = 4\ \text{k}\Omega$$

$$V_a = (2\ \text{mA})(4\ \text{k}\Omega) = 8.00\ \text{V}$$

$$V_b = \left(\frac{3\ \text{k}\Omega}{3\ \text{k}\Omega + 9\ \text{k}\Omega}\right)(8.0\ \text{V}) = 2.00\ \text{V}$$

b. After relocating the current source from the original location, and connecting it between node ***b*** and **ground**, we obtain the circuit shown:



The resulting node voltages are now found as follows:

$$R_T = 3\ \text{k}\Omega \| (6\ \text{k}\Omega + 9\ \text{k}\Omega) = 2.50\ \text{k}\Omega$$

$$V_b = (2\ \text{mA})(2.5\ \text{k}\Omega) = 5.00\ \text{V}$$

$$V_a = \left(\frac{6\ \text{k}\Omega}{6\ \text{k}\Omega + 9\ \text{k}\Omega}\right)(5.0\ \text{V}) = 2.00\ \text{V}$$

- **From the preceding results, we conclude that the reciprocity theorem again applies for the given circuit !!!**

- **Tutorials for enhancement:**

(1) Use Millman's theorem to find the current through and the power dissipated by R_L in the following circuits (Fig. a, b, c & d)

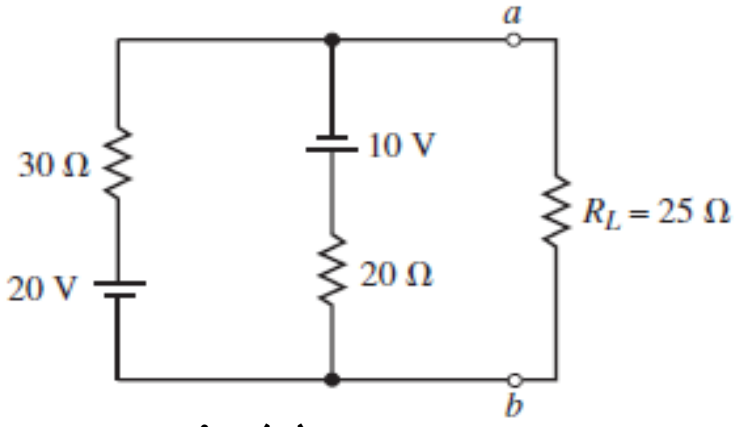


Fig. (a)

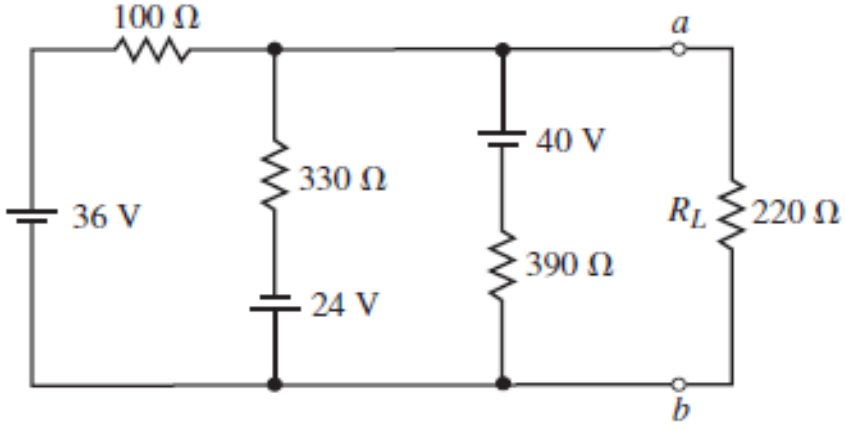


Fig. (b)

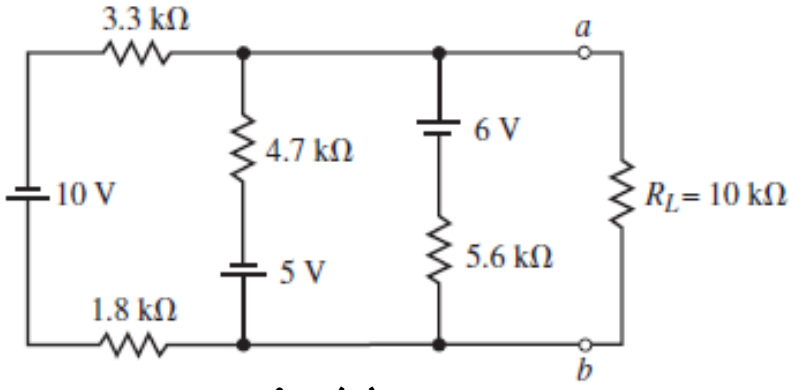


Fig. (c)

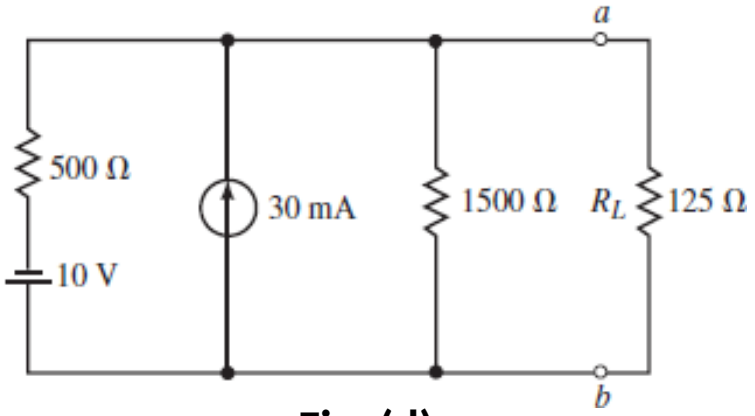


Fig. (d)

(2) –(a). Apply reciprocity theorem to determine the current *I* ***(Fig. a** & **b)** and the Voltage V **(Fig. c** & **d)*** in the given circuits.
(b) Show that reciprocity applies for the given circuit.

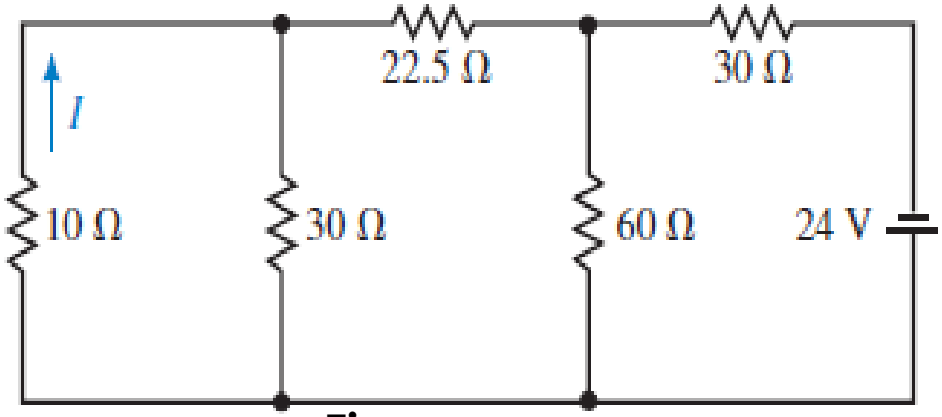


Fig. a

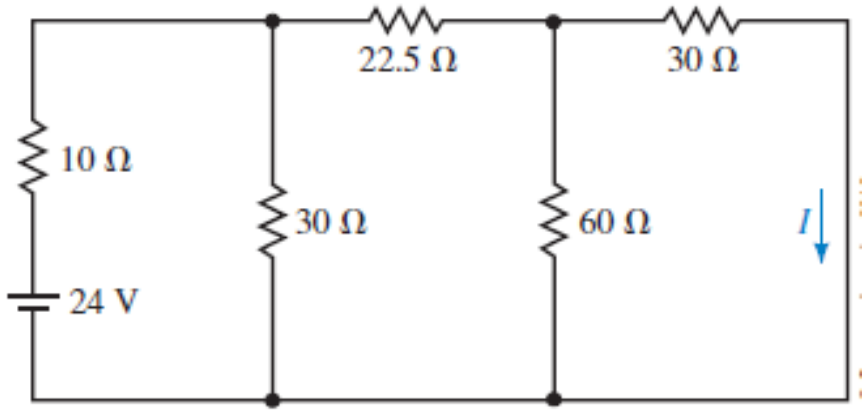


Fig. b

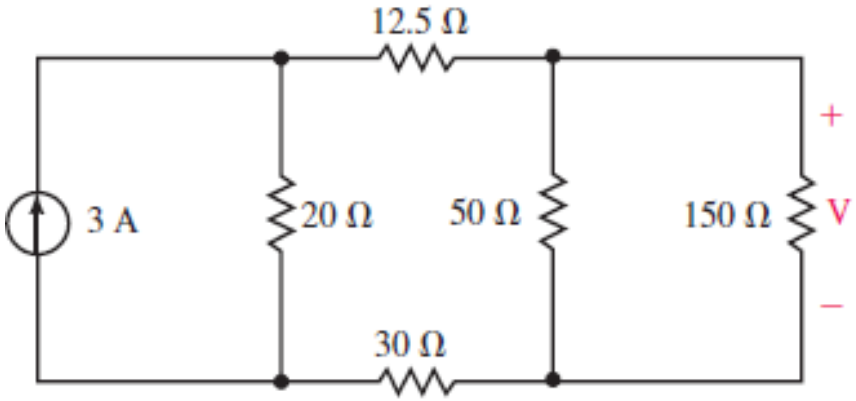


Fig. c

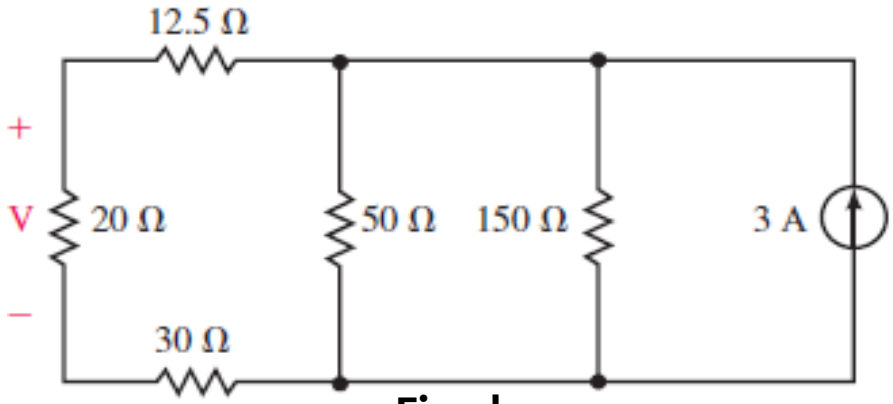


Fig. d

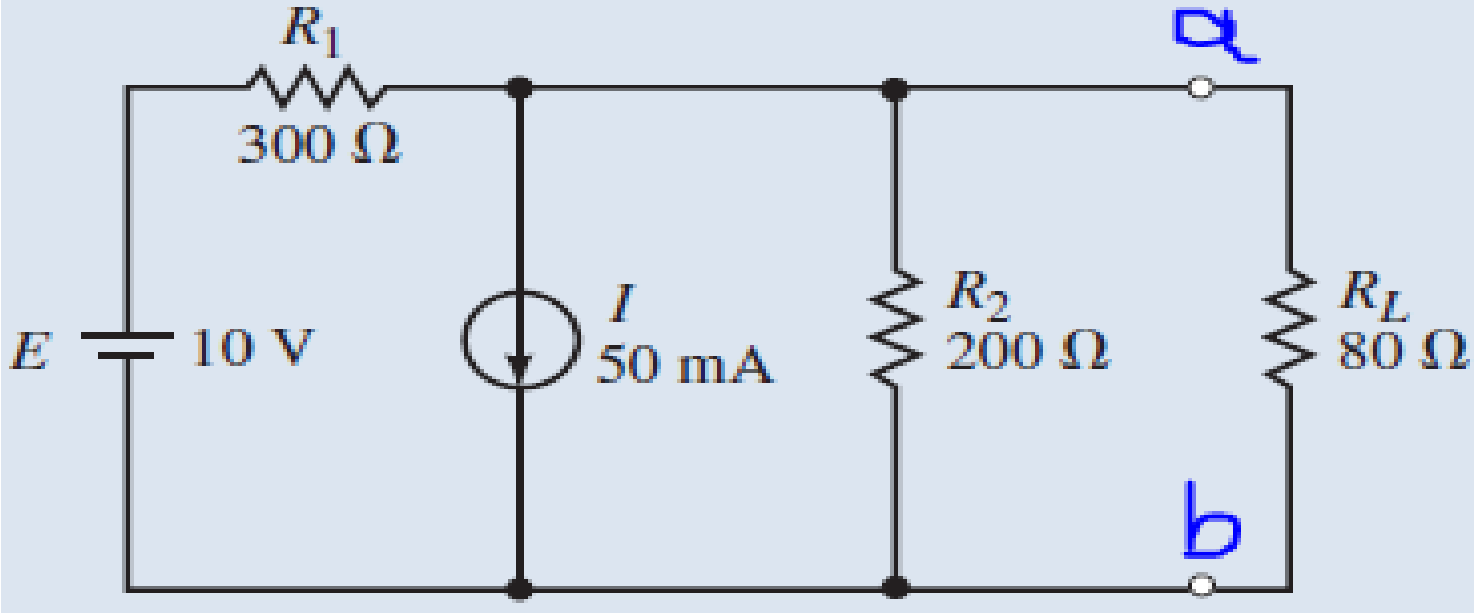
QUIZ

a) Use Thévenin's theorem to determine the current through load resistor R_L for the following Circuit/5.

b) Consider the Maximum Power transfer theorem and Calculate the Power that is supposed to be transferred to the load.

c) What is the efficiency ( ) at Maximum power?

d) What is the efficiency ( ) if the load is 100 ?



Assignment 1/10

By giving examples, do a detailed research and discuss the tellegen's theorem used in circuit network analysis